

4

Energy Sources and Energy Consumption

In this chapter, we will learn about

- non-renewable energy sources including fossil fuels and nuclear power
- renewable energy sources including wind, hydroelectric and solar power
- the environmental impact of energy consumption
- energy consumption patterns in Hong Kong

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4.1 Renewable and non-renewable energy sources

Flipped classroom
Watch a video and answer the questions.

Today, the world still generates electricity mainly from traditional fuels such as crude oil, coal and natural gas (Fig 4.1a). These fuels, known as **fossil fuels**, are **non-renewable energy sources** because once they are used up, they cannot be readily replaced. On the contrary, **renewable energy sources** are inexhaustible.

There are two main types of non-renewable energy sources, namely fossil fuels and **nuclear power**.

We usually use the term 'power' when energy is used to generate electricity.

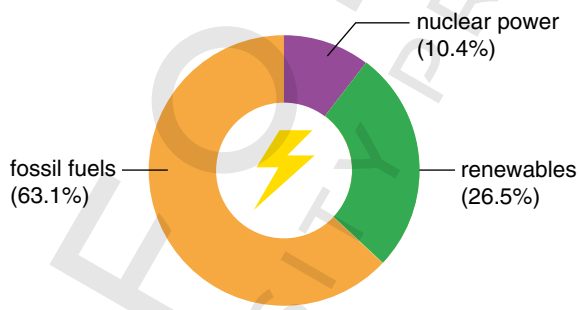


Fig 4.1a *World electricity production by source 2019. (Data obtained from IEA website in Aug 2022)*

DSE goal
Describe the characteristics of non-renewable energy sources and give examples.

1 Non-renewable energy sources

a Fossil fuels

Fossil fuels were formed naturally in the earth's crust from the remains of dead plants and animals over millions of years. Coal (Fig 4.1b) was formed from dead plants while crude oil (Fig 4.1c) and natural gas were mainly formed from dead marine organisms.



Fig 4.1b *Coal in an open-pit mine.*

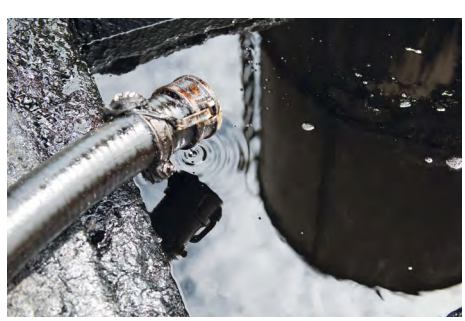


Fig 4.1c *Crude oil from oil well.*

Fossil fuels release energy by burning. They are heavily used for generating electricity and running vehicles.

Know more

Shale gas

Shale gas has become an increasingly important energy source in the past 20 years. Do you know what it is?

▸ p.185

fossil fuel 化石燃料　non-renewable energy source 不可再生能源
renewable energy source 可再生能源　nuclear power 核能　shale gas 頁岩氣

b Nuclear power

Nuclear power is the use of nuclear reactions to produce electricity. Presently, the vast majority of electricity from nuclear power stations is produced by nuclear fission of uranium and plutonium. The supply of such radioactive fuels is finite, therefore, nuclear power is considered non-renewable. We shall first study where the energy in a nuclear reaction comes from.

Know more

More about nuclear power

Nuclear power can also be obtained from nuclear decay and nuclear fusion reactions. A nuclear battery can use the energy released from nuclear decay to generate electricity. It is mainly used in isolated equipment such as spacecraft, pacemakers etc. At present, commercial fusion power station is still in development.

i Binding energy

DSE goal

Define binding energy per nucleon in unit of eV and solve problems.

Relate the binding energy curve to nuclear fission and fusion.

DSE exam

13(2)Q3.8, 14(2)Q3.7, 17(2)Q3.8, 18(2)Q3.8, 19(2)Q3.8, 20(2)Q.3(a)–(c), 21(2)Q3.8, 23(2)Q3.8

In Book 5 Chapter 3, we learned that fission releases nuclear energy when heavy nuclei are split, and fusion releases nuclear energy when light nuclei are joined. This nuclear energy is related to the **binding energy** of the nuclei.

The binding energy is not the energy that a nucleus possesses, but the energy required to break the nucleus into its constituent nucleons. It is also equal to the amount of energy released when protons and neutrons come together to form a nucleus (Fig 4.1d).

The following relation can be obtained.

Binding energy = separated nucleons – nucleus
(mass difference between nucleus and its constituent nucleons)

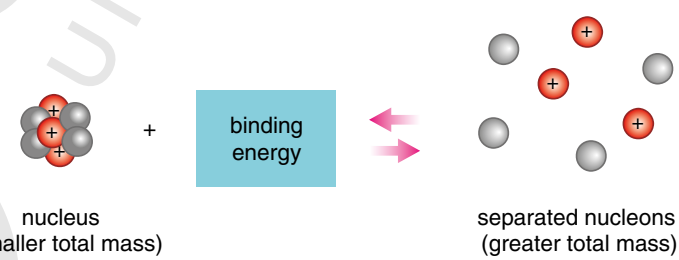


Fig 4.1d *The absorption and release of binding energy.*

The mass of a nucleus is less than the total mass of its constituent nucleons. This difference in mass, which is known as *mass defect*, is equivalent to the binding energy.

The binding energy of a nucleus is the amount of energy required to break a nucleus into its constituent nucleons.

binding energy 結合能　　mass defect 質量虧損

Einstein's mass-energy relationship can be used to find the binding energy E_b of a nucleus.

$$E_b \text{ (in J)} = \Delta m \text{ (in kg)} \times c^2$$

where Δm is the mass defect.

Cross-link
Refer to Book 5 Ch 3.2 for the definition of electron-volt (eV) and atomic mass unit (u).

▶ It is more common to express E_b in MeV. Since 1 u = 931 MeV, the binding energy is also given by

$$\boldsymbol{E_b \text{ (in MeV)} = \Delta m \text{ (in u)} \times 931}$$

The larger the nucleus, the greater the binding energy is. However, if we want to compare the stability of different nuclei, we need to calculate their **binding energy per nucleon** $\boldsymbol{\frac{E_b}{A}}$, where A is the mass number, i.e. the number of nucleons of the nucleus. **A nucleus with a larger** $\boldsymbol{\frac{E_b}{A}}$ **is more stable because its nucleons are more tightly bound.**

Example 1 Comparing stability of different nucleus

The mass of a carbon-12 (C-12) nucleus, a proton and a neutron in atomic mass unit (u) are given in the table.

C-12 nucleus	11.996 710 u
proton	1.007 276 u
neutron	1.008 665 u

(a) A C-12 nucleus has 6 protons and 6 neutrons. Calculate its binding energy and binding energy per nucleon in MeV.

(b) The mass defect of a uranium-234 (U-234) nucleus is 1.909 380 u. Explain whether the nucleus of C-12 or U-234 is more stable.

Solution

(a) $\Delta m = 6 \times 1.007\,276 + 6 \times 1.008\,665 - 11.996\,710 = 0.098\,936$ u

Binding energy = $0.098\,936 \times 931 = 92.1$ MeV

Binding energy per nucleon = $\frac{92.1}{12} = 7.68$ MeV

(b) Binding energy per nucleon of U-234 nucleus = $\frac{\Delta m \times 931}{A}$

$$= \frac{1.909\,380 \times 931}{234}$$

$$= 7.60 \text{ MeV}$$

This shows that the binding energy per nucleon of a more massive nucleus is not necessarily greater.

▶ C-12 nucleus is more stable because its binding energy per nucleon is larger.

Google Forms Microsoft Forms

Quick check 1

→ E-version in OUP Exercise Platform is also available.

1 Which of the following is/are non-renewable energy source(s)? Put a tick in the corresponding box(es).

- ☐ Crude oil
- ☐ Coal
- ☐ Natural gas
- ☐ Nuclear power

2 *True or false:*

(a) The energy released when a nucleus splits into smaller nuclei is the binding energy of the nucleus. (T/F)

(b) A larger nucleus has a larger binding energy. (T/F)

(c) A more stable nucleus has a greater binding energy per nucleon. (T/F)

3 A nucleus, with mass number A, has a binding energy E_b and a binding energy per nucleon $\frac{E_b}{A}$. The difference in mass between the nucleus and its constituent nucleons is Δm. Complete the table.

	Δm / u	E_b / MeV	A	$\frac{E_b}{A}$ / MeV
(a)	0.600 530		64	
(b)		1830	243	
(c)			56	8.8

Learn by practice 1

1 The mass of an oxygen-16 nucleus is smaller than the total mass of its constituent nucleons by 0.137 005 u. The mass of a tungsten-180 nucleus is smaller than the total mass of its constituent nucleons by 1.551 651 u. Explain which of the two nuclei is more stable.

2 The binding energy of $^{92}_{40}\text{Zr}$ is 799.722 107 MeV.

(a) How many protons and neutrons are there in a $^{92}_{40}\text{Zr}$ nucleus?

(b) Find the mass of a $^{92}_{40}\text{Zr}$ nucleus.

Given: mass of proton = 1.007 276 u
mass of neutron = 1.008 665 u

In nuclear fission and fusion, the energy released is equal to the increase in binding energy in the reaction. Let us look at the following example.

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Example 2 Energy released in a fission reaction

The following equation shows one possibility when a nucleus of uranium-235 absorbs a neutron and splits into lighter nuclei.

$$^{1}_{0}\text{n} + {}^{235}_{92}\text{U} \rightarrow {}^{144}_{56}\text{Ba} + {}^{89}_{36}\text{Kr} + 3{}^{1}_{0}\text{n}$$

The table below shows the binding energy per nucleon of the particles.

Particle	Binding energy per nucleon / MeV
$^{235}_{92}$U	7.590 937
$^{144}_{56}$Ba	8.265 550
$^{89}_{36}$Kr	8.616 942

(a) Find the energy released in the reaction in (i) MeV and (ii) J.

(b) State the change in the total mass of particles in the reaction.

Solution

(a) (i) Binding energy of $^{235}_{92}$U = 7.590 937 × 235 = 1783.87 MeV

Binding energy of $^{144}_{56}$Ba = 8.265 550 × 144 = 1190.24 MeV

Binding energy of $^{89}_{36}$Kr = 8.616 942 × 89 = 766.91 MeV

Energy released = 1190.24 + 766.91 − 1783.87 = 173 MeV

(ii) Energy released = $173 \times 10^{6} \times 1.60 \times 10^{-19} = 2.77 \times 10^{-11}$ J

(b) Decreases

Tip

1 eV = 1.60×10^{-19} J

Since energy is released in the reaction, from $E = mc^2$, the total mass of the reaction products should be smaller than that of the particles before the reaction.

Google Forms Microsoft Forms

Quick check 2

→ E-version in OUP Exercise Platform is also available.

1 A nuclide X undergoes fission to form two nuclides Y. Energy is released in the reaction. Determine whether the following statements about X are true.

(a) The total binding energy of X is greater than that of Y. (T/F)

(b) The binding energy per nucleon of X is smaller than that of Y. (T/F)

(c) The mass of X is larger than double the mass of Y. (T/F)

Learn by practice 2

1 The equation below represents a fusion reaction. Find the energy released in the reaction in (a) MeV and (b) J.

$$^{2}_{1}\text{H} + {}^{3}_{1}\text{H} \rightarrow {}^{4}_{2}\text{He} + {}^{1}_{0}\text{n}$$

Given: binding energy per nucleon of $^{2}_{1}$H nucleus = 1.112 283 MeV

binding energy per nucleon of $^{3}_{1}$H nucleus = 2.827 266 MeV

binding energy per nucleon of $^{4}_{2}$He nucleus = 7.073 915 MeV

When triggered, less stable nuclei are more likely to undergo nuclear reactions, forming more stable nuclei. Let us have a look on the binding energy curve where $\frac{E_b}{A}$ is plotted against A for different nuclides (Fig 4.1e).

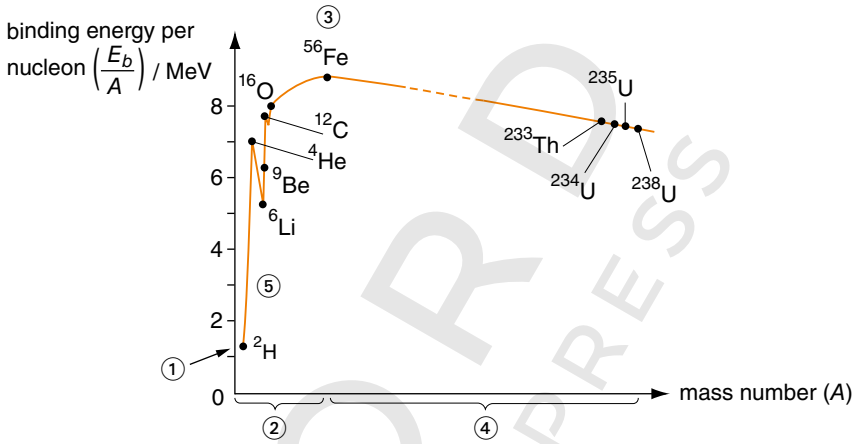


Fig 4.1e *Binding energy curve.*

The binding energy curve has the following features:

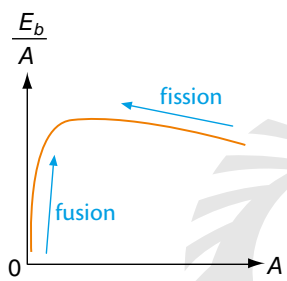
① The graph line does not start at $A = 0$. At least two nucleons are needed for binding energy to exist.

② In this region, $\frac{E_b}{A}$ generally increases with A. Therefore, it is favourable for light nuclides to undergo fusion to form more stable nuclides and release energy.

Nickel-62 has the highest binding energy per nucleon, followed by iron-58 and iron-56.

This is called the iron limit. Fusion reactions are no longer energetically favourable in producing nuclides heavier than iron.

③ Iron-56 ($^{56}_{26}$Fe) is at the peak. It is a very stable nuclide. If it undergoes fission or fusion, the resultant nuclides will be less stable and energy will be absorbed rather than released.



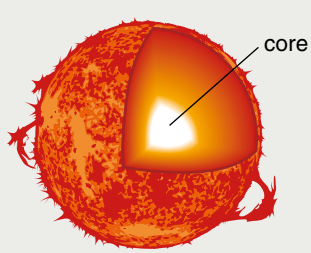
④ In this region, $\frac{E_b}{A}$ decreases when A increases. Therefore, it is favourable for heavy nuclides to undergo fission to form more stable nuclides and release energy.

⑤ The slope in region ② is steeper than in region ④. This indicates that fusion in general produces more energy per nucleon than fission.

Know more

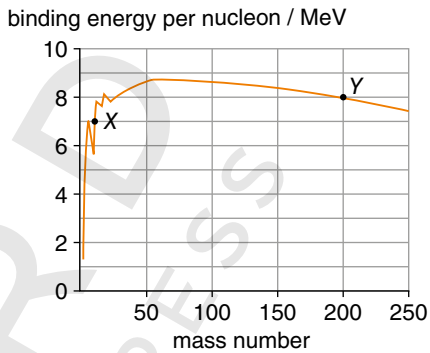
Fusion in stars

The energy of a star comes from nuclear fusion in its core. At first, hydrogen fuses to form helium. Then helium fuses to form carbon. In massive stars, fusion continues to form heavier elements. The process stops after producing iron since energy will no longer be released by fusion.



Example 3 Determine type of reaction from binding energy curve

X and Y are two different nuclides. Their positions on the binding energy curve are shown on the right.



(a) Estimate the binding energy of Y from the curve.

(b) Which nuclide is more likely to undergo fusion? Explain.

Solution

(a) Binding energy = $8 \times 200 = 1600$ MeV

(b) X is more likely to undergo fusion. This is because if X undergoes fusion, the nuclide formed will have higher binding energy per nucleon and is more stable. On the other hand, if Y undergoes fusion, the nuclide formed will have lower binding energy per nucleon and is less stable.

Tip

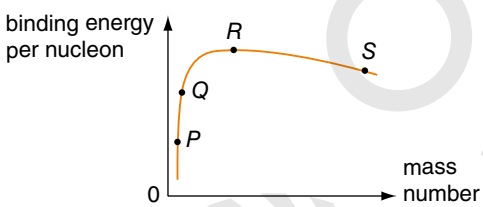
From the binding energy curve, $\frac{E_b}{A} = 8$, $A = 200$.

Google Forms Microsoft Forms

Quick check 3

→ E-version in OUP Exercise Platform is also available.

1 Consider the binding energy curve below. P, Q, R and S correspond to different nuclides.



Determine whether the following statements are true.

(a) If nuclear energy were to be generated by the fusion of one of the nuclides, the nuclide is most likely to be S. (T/F)

(b) R is the most stable among the four nuclides. (T/F)

(c) Q is more stable than S. (T/F)

Learn by practice 3

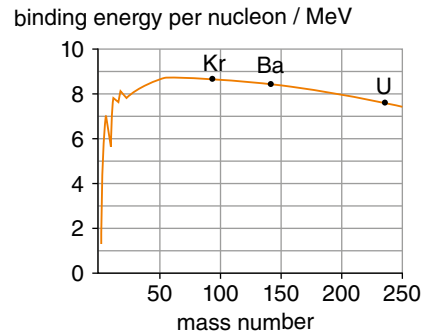
1 In the following nuclear fission reaction, 173.272 114 MeV of energy is released.

$$^{235}_{92}\text{U} + {}^{1}_{0}\text{n} \rightarrow {}^{141}_{56}\text{Ba} + {}^{92}_{36}\text{Kr} + 3{}^{1}_{0}\text{n}$$

The binding energy per nucleon of $^{141}_{56}$Ba and $^{92}_{36}$Kr nuclei are 8.326 028 MeV and 8.512 845 MeV respectively.

(a) Find the binding energy per nucleon of a $^{235}_{92}$U nucleus.

(b) Referring to the binding energy curve on the right, will the fission products tend to undergo fission or fusion reaction?



DSE goal

Describe the principle of the fission reactor and state the roles of moderator, coolant and control rods.

DSE exam

12(2)Q3.8, 14(2)Q3.8, 15(2)Q3.8, 20(2)Q.3(d)

ii Fission reactor

Some nuclear power stations make use of the fission of plutonium-239 to generate electricity. A uranium-238 nucleus in the fuel captures a neutron to become a plutonium-239 nucleus.

Many nuclear power stations make use of the fission of uranium-235 (U-235) to generate electricity. A chain reaction is essential and the process is as follows:

① A U-235 nucleus splits into two lighter nuclei after capturing a **slow-moving neutron** (Fig 4.1f).

② More neutrons and a huge amount of energy are released.

③ The released neutrons continue splitting other U-235 nuclei, causing a chain reaction.

The nuclear fuel are made of enriched uranium oxide. The percent composition of uranium-235 in the fuel elements is about 3–4% while that in natural uranium is about 0.7%.

Cross-link

Refer to Book 5 Ch 3 for the details of nuclear fission and chain reaction.

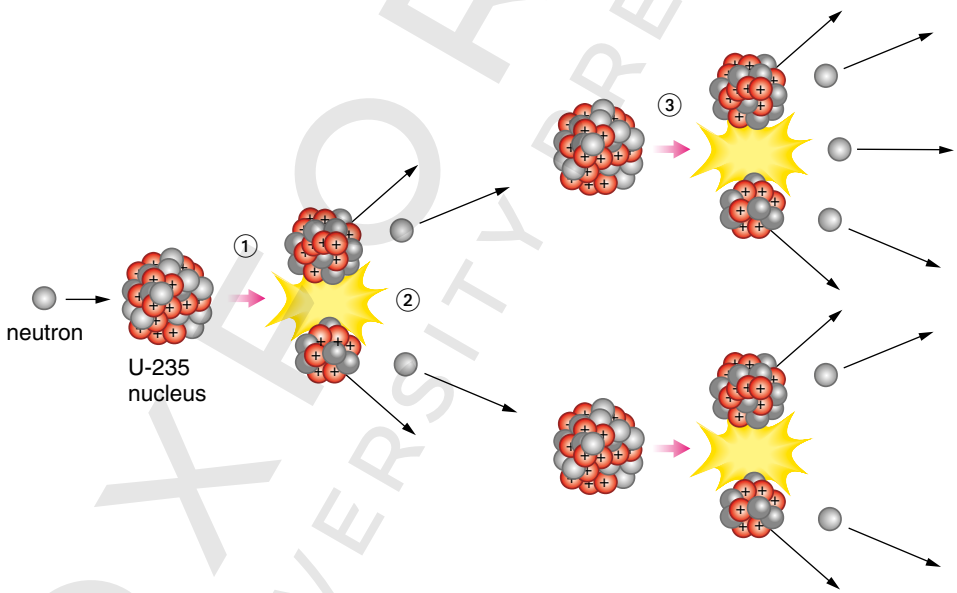


Fig 4.1f Chain reaction of uranium-235.

In a nuclear power station, nuclear fission takes place in a reactor where the rate of fission can be precisely controlled. There are different types of nuclear reactors, but their working principles are of certain similarities. In the following, we will use a **pressurized water reactor**, or PWR for short, as an example for our discussion (Fig 4.1g).

The nuclear power stations at Daya Bay use PWRs.

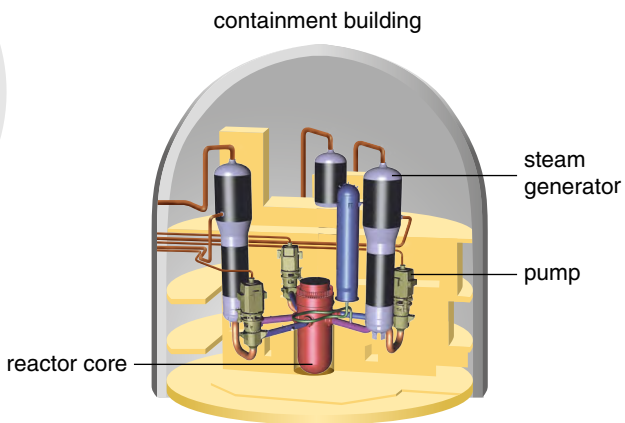


Fig 4.1g Structure of a PWR.

pressurized water reactor 壓水式反應堆

Simulation

→ This simulation shows how a pressurized water reactor works. It is available on OUP web in PC version (.exe) only.

Let us take a look at how a power station employing a PWR works (Fig 4.1h).

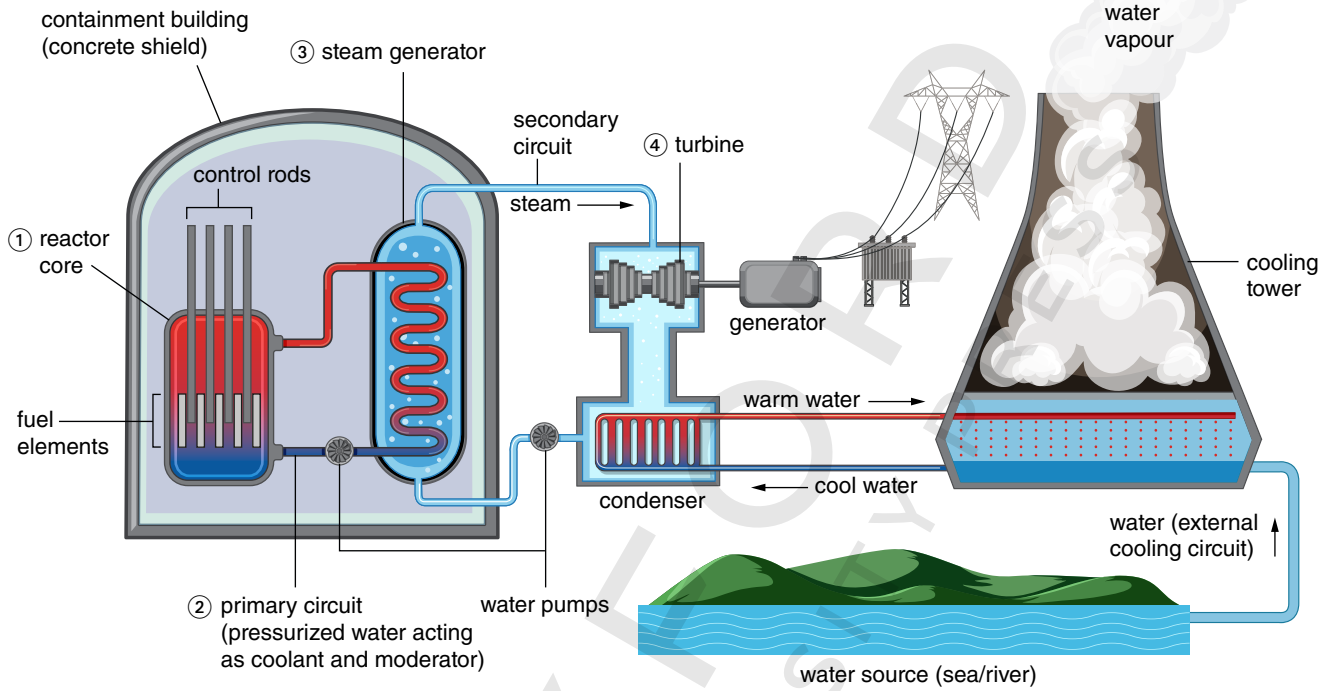


Fig 4.1h *Components of a power station employing a PWR.*

① Inside the reactor core, fission reactions occur in the fuel elements. The energy released is mostly converted into the kinetic energy of the fission products.

When the fission products collide with water molecules, part of the KE becomes internal energy of the water.

② In the primary circuit, pressurized water is pumped through the reactor core. The water acts as

- the **coolant** to **absorb the energy** released by the fission reactions and transfers it to the steam generator;

The fuel elements may melt if the coolant fails to function effectively. ▶

- the **moderator** to **slows down the fast-moving neutrons** produced by fission reactions, thus maintaining the chain reaction.

If the neutrons move too fast, they are unlikely to trigger other fission reactions and the chain reaction will stop. ▶

③ The hot pressurized water passes through the steam generator. The water in the secondary circuit is heated and vaporized.

④ The steam passes through turbines that drive a generator to produce electricity. An external cooling circuit is used to turn the steam back into water to complete the cycle in the secondary circuit.

The concentration of the fissionable nuclei decreases over time. When the concentration is too low, the chain reaction cannot be sustained. Therefore, the nuclear fuel is usually replaced every 12 to 24 months even it is not used up.

✎ Ts may ask Ss to explore a PWR using the interactive model found at the following website. Note that it takes time to load the model.

coolant 冷卻劑　　moderator 減速劑

A power station employing a PWR has the following safety features:

Control rods are usually made of boron steel. They are inserted into the reactor core from above. To shut down a reactor urgently, the control rods will be allowed to fall, under gravity, completely into the reactor core.

1 **Control rods** are used to **absorb neutrons in order to reduce the rate of fission** (Fig 4.1i). When the reaction is too fast, the control rods are inserted deeper to absorb more neutrons. In an emergency, the control rods are completely inserted into the core to stop the chain reactions.



Fig 4.1i *Control rods inserted into reactor cores.*

The pressurized water in the primary circuit is in contact with the fuel elements and contains radioactive substances.

2 **Water in the primary and secondary circuits does not mix to prevent contamination**. Otherwise, the radioactive substances in the primary circuit may get into the water in the secondary circuit and contaminate the environment.

3 If there is a leak in the reactor core, **the concrete containment building helps to prevent radioactive substances from contaminating the environment**.

Right and wrong

✗ In a fission reactor, control rods act as the moderator.

✓ In a fission reactor, pressurized water acts as the moderator to slow down neutrons so that fission can be maintained at a high enough rate.

Google Forms Microsoft Forms

Quick check 4

→ E-version in OUP Exercise Platform is also available.

1 *True or false:*

(a) If the control rods are completely inserted in the reaction core of a nuclear fission reactor, the chain reaction will stop. (T/F)

(b) In a pressurized water reactor, the water in the secondary circuit is radioactive. (T/F)

(c) The moderator in a nuclear fission reactor can absorb neutrons and slow down the rate of fission. (T/F)

2 The following steps are about generating electricity in a power station employing a PWR. Put the steps in the correct order.

(1) The turbines drive the generator.

(2) The hot water passes through the steam generator to produce steam.

(3) The steam drives the turbines.

(4) The chain reaction of fission occurs.

(5) The pressurized water is heated.

Know more

Pressurized water

The pressurized water is under a pressure of about 153 atm (15.5 MPa). It has a higher boiling point than regular water. It can be heated to above 300 °C without boiling after passing through the reactor core. Energy can then be transferred to the steam generator at a high rate.

control rod 控制棒

Know more

Self-regulating

The use of water as a moderator is also a safety feature of a PWR. If the temperature of water becomes higher than normal, it will be less effective in slowing down the neutrons, causing the rate of fission reactions to decrease. This process is referred to as 'self-regulating'.

Site selection of nuclear power station

Nuclear power stations need a large amount of water to operate, so they are usually built by the sea or river to ensure the water supply.



A nuclear power station by the sea.

Residual heat

When a reactor is shut down, all fission reactions stop. However, the fuel elements still generate heat because the fuel and the fission products are radioactive and energy is emitted when they decay. This is called *residual heat*.

In the 2011 *Fukushima* nuclear accident, the reactors were shut down immediately after the earthquake. However, the tsunami destroyed the power supply and the cooling system stopped working as a result. The residual heat melted the fuel elements and the reactor cores, causing highly radioactive substances to leak out, and spread by water to the surrounding area.



Satellite image of the Fukushima nuclear power station taken after the 2011 earthquake.

residual heat 餘熱　　Fukushima 福島

☆ HKDSE 2020 Paper 2 Q.3

→ This video demonstrates how this question is solved.

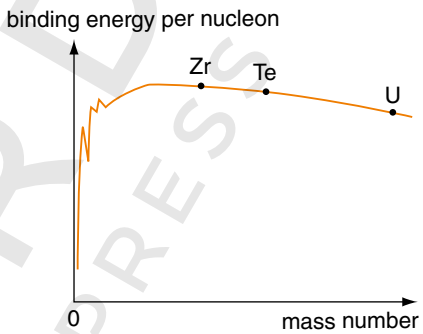
Video

DSE practice 1 | Power output of a nuclear power station

The following reaction of uranium-235 (U-235) takes place in a nuclear reactor.

$$^{235}_{92}\text{U} + ^{1}_{0}\text{n} \rightarrow ^{94}_{40}\text{Zr} + ^{139}_{52}\text{Te} + 3^{1}_{0}\text{n}$$

(a) Refer to the binding energy curve on the right. Explain whether energy is absorbed or released in the above reaction. (2 marks)



(b) The binding energy of the particles involved in the reaction are shown below.

Particle	Binding energy / MeV
$^{235}_{92}\text{U}$	1783
$^{139}_{52}\text{Te}$	1142
$^{94}_{40}\text{Zr}$	814.7

The energy released in the reaction contributes to 97.5% of the energy released by the nuclear reactor. The efficiency of the nuclear power station in converting the energy released by the nuclear reactor to electrical energy is 35%. Suppose 0.04 g of U-235 are used every second to release energy through the above reaction in the nuclear reactor.

(i) Find the energy released in the above reaction in J. (1 mark)

(ii) Find the number of U-235 nuclei undergoing the above reaction per second.
Given: molar mass of $^{235}_{92}\text{U}$ = 235 g (1 mark)

(iii) Estimate the power output of the power station in MW. (2 marks)

(c) Control rods are components in a fission reactor. Describe their functions. (2 marks)

Solution

(a) Energy is released in the fission as the fission products $^{94}_{40}\text{Zr}$ 1A
and $^{139}_{52}\text{Te}$ have higher binding energy per nucleon than $^{235}_{92}\text{U}$. 1A

(b) (i) Energy released = $[(1142 + 814.7) - 1783] \times 10^{6} \times 1.60 \times 10^{-19}$
$= 2.779 \times 10^{-11}$ J $\approx 2.78 \times 10^{-11}$ J 1A

(ii) Number of U-235 nuclei = $\frac{0.04}{235} \times 6.02 \times 10^{23}$
$= 1.0247 \times 10^{20} \approx 1.02 \times 10^{20}$ 1A

(iii) Power output of the power station
$= 2.779 \times 10^{-11} \times 1.0247 \times 10^{20} \div 97.5\% \times 35\%$ 1M
= 1020 MW 1A

Common mistake
Students may make mistakes in unit conversion or in manipulating the efficiency of energy conversion.

(c) The control rods control the rate of the fission reaction by absorbing neutrons and 1A
stop the reaction in case of emergency. 1A

Common mistake
Students may fail to state clearly that the control rods control the rate of reaction.

▶ Revision 4 Q30 (p.183)

DSE goal

Describe the characteristics of renewable energy sources and give examples.

DSE exam

12(2)Q.3(d), 20(2)Q3.8

2 Renewable energy sources

Renewable energy sources refer to energy sources that are available in unlimited quantities. There are many kinds of renewable energy. We shall focus on **wind power**, **hydroelectric power** and **solar power**. They are regarded as sustainable because they produce relatively little pollution and so have a low environmental impact.

a Wind power

DSE goal

Derive maximum power by wind turbine as $P = \frac{1}{2}\eta\rho Av^3$, where η is the efficiency and solve problems.

DSE exam

13(2)Q3.4, 14(2)Q3.5, 17(2)Q3.6, 18(2)Q3.7, 19(2)Q3.7, 21(2)Q3.2, 22(2)Q3.8, 23(2)Q3.8, etc.

✎ In Hong Kong, about 0.4% of electricity comes from renewable energy. The major types of renewable energy available in Hong Kong are solar energy, wind energy and biogas.

Wind energy is the energy obtained from the kinetic energy of air. It has been used by humans for thousands of years for sailing, grinding grains and pumping water (Fig 4.1j). Today, **wind turbines** (Fig 4.1k) are used to convert wind energy into electrical energy. Wind power is one of the fastest-growing renewable energy resources.



Fig 4.1j *Traditional windmills*



Fig 4.1k *Modern wind turbines*

The working principle of a wind turbine is simple: the wind drives the *rotor* (i.e. the rotating part of the turbine, including the *blades*), which in turn drive a generator to produce electricity.

The energy conversion in a wind turbine is as follows:

KE of air → KE of rotor → electrical energy

Consider a wind blowing normally with a constant speed v onto the rotor blades of a wind turbine. The blades sweep a circular region of area A. In a time interval t, all the air passing through the circular region is in the cylinder shown in Figure 4.1l.

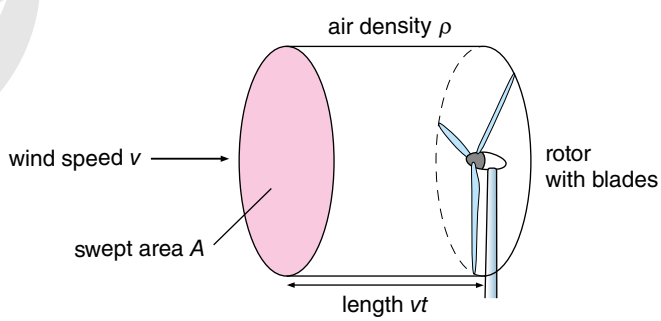


Fig 4.1l *Air passing through the rotor of a wind turbine in time interval t.*

wind power 風能　hydroelectric power 水力發電　solar power 太陽能　wind turbine 風力渦輪機
rotor 轉子　blade 扇葉

From Figure 4.1l, we can see that

volume of air passing through the circular region = $A \times vt$

Take the density of air as ρ kg m^{-3}.

Density = $\frac{\text{mass}}{\text{volume}}$

mass = density × volume

Mass of air passing through the circular region = $\rho \times Avt$

Total KE of air before passing through the circular region

$$= \frac{1}{2}mv^2$$

$$= \frac{1}{2}(\rho Avt)v^2$$

$$= \frac{1}{2}\rho Av^3t$$

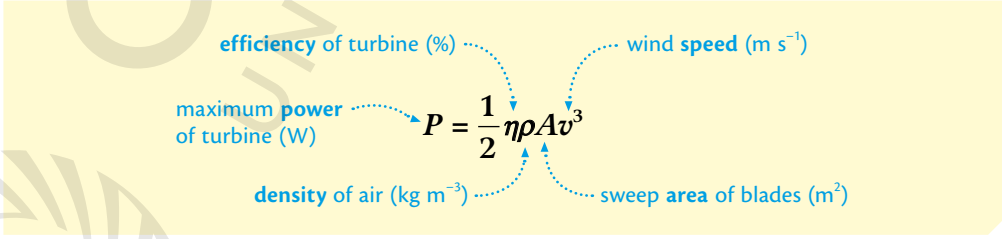
which is the maximum energy input from the air.

The rotor produces noise as it turns, and so some energy is also lost as sound. That is why wind turbines are often located far away from residential areas.

A wind turbine cannot convert all this energy into electricity because there is energy loss, such as the work done against the friction between the moving parts. Also, air does not stop after passing through the rotor, so not all its KE is transferred to the rotor. Therefore, the efficiency η of a wind turbine in converting the KE of air into electrical energy must not achieve 100%. The **maximum** power output P is given by

$$P = \frac{\text{maximum energy input from the air}}{\text{time}} \times \text{efficiency}$$

$$= \frac{\frac{1}{2}\rho Av^3t}{t} \times \eta$$



The maximum power P turbine varies directly with v^3, which means doubling the wind speed results in eight times as much power.

The actual power generated is usually below this maximum value because the wind direction is not always perpendicular to the rotor.

Know more

Structure of a wind turbine

What are the components inside a wind turbine? How do they make a wind turbine work?

▸ p.184

Example 4 Power of Lamma Winds

In *Lamma Winds*, the blades of the wind turbine are 25 m long. A wind blows with a constant speed of 5.56 m s^{-1}. Take the efficiency of the turbine as 35%, and the air density as 1.225 kg m^{-3}.



(a) Find the maximum power generated by the wind turbine in kW.

(b) If the wind speed increases by 50%, what will the maximum power become? Assume the efficiency of the turbine remains the same.

The average wind speed in Hong Kong is 20 km h^{-1} (5.56 m s^{-1}).

Solution

(a) Maximum power $= \frac{1}{2}\eta\rho A v^3$

$$= \frac{1}{2} \times 35\% \times 1.225 \times (\pi \times 25^2) \times 5.56^3$$

$$= 72.3 \text{ kW}$$

(b) As $P \propto v^3$, if v increases by 50%,

maximum power $= 72.3 \times (1 + 50\%)^3 = 244$ kW

In practice, the **efficiency** of a turbine **changes at different wind speeds**. Especially when the wind speed is too low, the wind is not able to overcome the friction between the moving parts. The blades are not turned and so no power is generated. On the other hand, when the wind speed is too high, the turbine will be stopped automatically to prevent damage. Hence, the power output also falls to zero.

The wind speed limit for a turbine to shut down because of spinning too fast is known as the cut-out speed. The value varies by turbine. The cut-off speed for Lamma Winds is 25 m s^{-1}.

STSE

Renewable energy sources in Hong Kong

Lamma Winds is the only large-scale wind power station in Hong Kong. Do you want to know more about this station as well as other renewable energy sources in Hong Kong?

▶ p.186

Know more

Offshore wind farm in Hong Kong

Land is a valuable resource. Building a wind farm offshore can avoid taking up precious land space. Also, the wind at sea is more stable and more powerful. Offshore wind speed is about 90% higher on average than onshore wind speed.

HK Electric planned to build an offshore wind farm at about 4 km away from the Lamma Power Plant, while CLP Power proposed to build one in the east of the Clear Water Bay Peninsula. Both had planned to put into operation in 2027. However, the plans were postponed due to the high overall cost and power generation cost.

Lamma Winds 南丫風采發電站

Google Forms Microsoft Forms

Quick check 5

→ E-version in OUP Exercise Platform is also available.

1 Consider a wind turbine of output power P operating at wind speed v. Assume wind always blows normally to the rotor blades and the efficiency remains fixed for all wind speeds. Determine whether the following statements are true.

(a) Its power output is $2P$ if the lengths of the blades are doubled. (T/F)

(b) Its power output is $2P$ if wind speed is $2v$. (T/F)

2 The blades of a wind turbine are of length l and sweep an area A. The efficiency of the turbine is η. The air density is ρ. When the wind blows onto the blades at speed v, the maximum power generated by the wind turbine is P. Complete the table below.

	P / W	ρ / kg m^{-3}	v / m s^{-1}	l / m	A / m^2	η
(a)		1.2	15	35		35%
(b)	20 000	1.2		5		25%
(c)	900 000	1.23	13			30%

3 Which of the following explain(s) that the electrical energy generated by a wind turbine is always less than the kinetic energy of air reaching its rotor? Put a tick in the corresponding box(es).

☐ Wind does not stop completely after passing through the rotor.

☐ There are mechanical energy losses in the moving parts.

☐ The direction of wind changes irregularly.

Learn by practice 4

Take air density = 1.225 kg m^{-3}.

1 Find the maximum power generated by the wind turbine of Lamma Winds when the wind speed is 20 m s^{-1}. Take the length of the rotor blade as 25 m and the efficiency of the turbine as 10%.

2 Wind is blowing normally at 16 m s^{-1} at a wind turbine with rotor diameter of 40 m.

(a) If the turbine generates an electrical power output of 750 kW, what is the efficiency of the turbine at such wind speed?

(b) If the rotor diameter becomes 60 m and the efficiency is not affected, how many such wind turbine are required to generate an electrical power output of 100 MW (1 MW = 1×10^6 W) under the given wind condition?

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DSE goal

Describe the energy conversion process for hydroelectric power and solve problems.

DSE exam

12(2)Q3.7, 15(2)Q3.7, 16(2)Q.3, 17(2)Q3.7, 21(2)Q3.7, 22(2)Q3.3

b Hydroelectric power

Hydroelectric power works by using the energy of moving water. In most hydroelectric power stations, dammed water is used (Fig 4.1m) and the efficiency of the stations can be as high as 90%. Let us study the structure and the working principle of a hydroelectric power station (Fig 4.1n).



Fig 4.1m *The three Gorges Dam.*

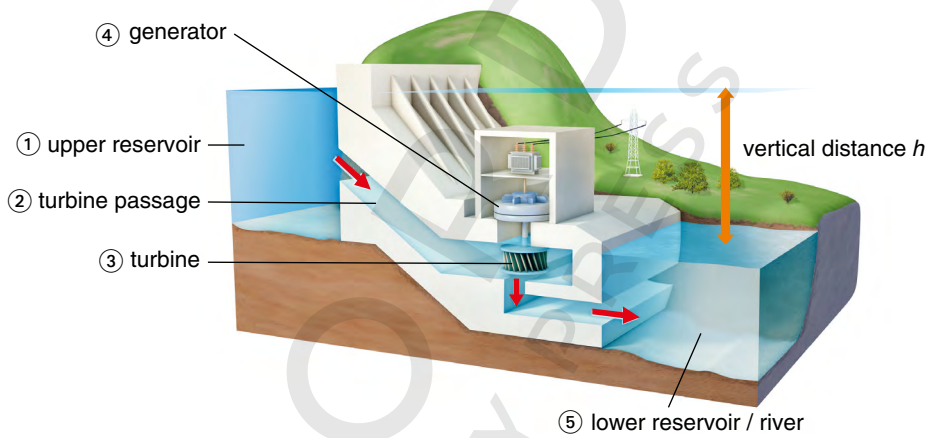


Fig 4.1n *The structure of a hydroelectric power station.*

① The water in the upper reservoir stores a huge amount of potential energy.

② When the turbine passage opens, water from the upper reservoir flows through the dam at a high speed.

③ The turbine in the passage is turned by the flowing water.

④ The generator produces electricity as the turbine spins.

⑤ Water is discharged to the lower reservoir or river.

The energy conversion in a hydroelectric power station is as follows:

PE of water → KE of water → KE of turbine → electrical energy

Let us estimate the electrical power generated by a hydroelectric power station. Suppose that the **vertical distance** between the water level at the upper reservoir and the lower reservoir is h. A turbine is located below the water level of the lower reservoir. Water of mass m flows through the turbine in a time interval of t. The efficiency of the power station is η.

$$\text{Power output} = \frac{\text{loss in PE of water}}{\text{time taken}} \times \text{efficiency}$$

Caution

The distance between the turbine and the water level of the lower reservoir does not affect the power output.

Usually power stations with larger dams have higher efficiency because the flow of water is more consistent.

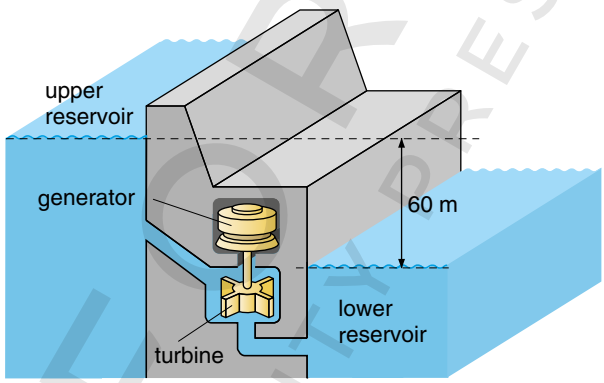
acceleration due to gravity (m s^{-2})
mass of water (kg)
power output (W)
time interval (s)

$$P = \frac{mgh}{t} \times \eta$$

vertical distance between upper and lower reservoirs (m)
efficiency of power station (%)

The efficiency η accounts for the energy loss due to the friction between the moving parts of the turbine as well as the resisting force acting on the water when the water passes through the turbine passage. Also, the water does not stop after passing through the turbine and does not transfer all its KE to the turbine.

Example 5 Efficiency of a hydroelectric power station



In a hydroelectric power station, the water level of the upper reservoir is 60 m above that of the lower one. Water flows from the upper to the lower reservoir at a rate of 5 m^3 s^{-1} and turns a turbine below the water level of the lower reservoir. The turbine is connected to a generator.

Given: density of water = 1000 kg m^{-3}, g = 9.81 m s^{-2}

1 MW = 1 × 10^6 W ▶

(a) Find the rate of loss of the potential energy of the water in MW.

(b) The output power of the station is 2.5 MW. Find its efficiency.

Solution

(a) Rate of loss of PE of water

$$= \frac{mgh}{t} = \left(\frac{m}{t}\right)gh = (5 \times 1000) \times 9.81 \times 60 = 2.94 \text{ MW}$$

(b) $$\text{Efficiency} = \frac{\text{power output}}{\text{power input}} \times 100\% = \frac{2.5}{2.94} \times 100\% = 85.0\%$$

Know more

Three Gorges hydropower station

The *Three Gorges hydropower station* on the *Yangtze River* in central China is the world's largest hydroelectric power station. It has generated a world record high electricity of 111.8 billion kW h in 2020.

Tidal power

Tidal power also makes use of water to generate electricity.

▶ p.185

Research also found that substantial amount of methane (a type of greenhouse gas) may be produced at the bottom of the dam.

A dam is often built to block water in a river and land is flooded to form a reservoir. ▶

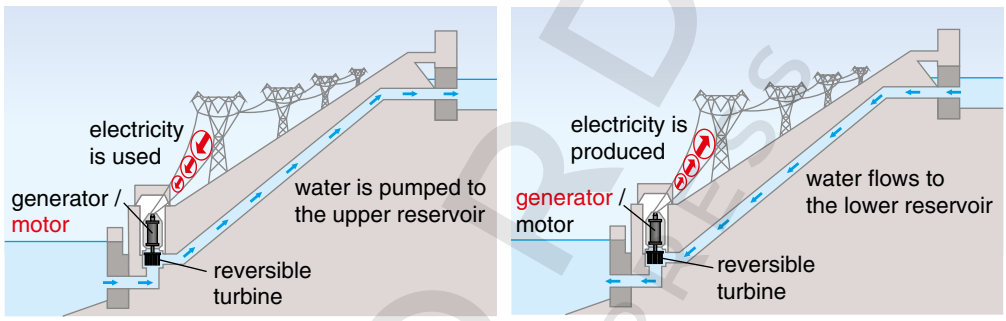
The output of hydroelectric power is steady and can be controlled. Besides, the cost of electricity is the lowest among all renewable energy sources.

Although hydroelectric power does not pollute the water or the air, its facilities have large environmental impacts by changing the natural habitats in the dam area.

Three Gorges hydropower station 三峽水電站 Yangtze River 長江 tidal power 潮汐能

When water flows from the upper to the lower reservoir through a turbine, electricity is produced. This process can be reversed. Water can be pumped from the lower to the upper reservoir using excess electricity from another power station at low-demand periods. Energy is stored in the form of PE of water and could be converted back to electricity at high-demand periods. This is called the **pumped hydroelectric storage system** (Fig 4.1o).

The upper reservoir of a storage system is usually small compared to the one of a hydroelectric power station. ▶



(i) Pumping water to upper reservoir (during low demand period).

(ii) Generating electricity (during high demand period).

Fig 4.1o *Working principle of a pumped hydroelectric storage system.*

Google Forms Microsoft Forms

Quick check 6

→ E-version in OUP Exercise Platform is also available.

1 Which of the following explain(s) that the electrical energy generated by a hydroelectric power station is always less than the change in PE of the water flowing from the upper reservoir to the lower reservoir? Put a tick in the corresponding box(es).

- ☐ The water does not stop moving after passing through the turbine.
- ☐ There is friction between the moving parts in the turbine.
- ☐ There is resistive force when water flows through the passage.
- ☐ The water level of the upper reservoir varies from time to time.

2 In a hydroelectric power station, water flows from the upper reservoir at a rate of $\frac{m}{t}$ to turn turbines below the water level of the lower reservoir. The vertical distance between the water levels of the reservoirs is h. The efficiency of the station is η. The power output of the station is P. Complete the table below.

	P / W	$\frac{m}{t}$ / kg s^{-1}	h / m	η
(a)		24 000	180	85%
(b)	5×10^9		150	75%
(c)	1×10^6	3000	50	

Learn by practice 5

1 The difference in water levels of a hydroelectric power station is 55 m. The efficiency of the station in electricity generation is 60%. If the flow rate of the water is 510 m^3 s^{-1}, what is the power output of the station in MW?
Given: density of water = 1000 kg m^{-3}

2 A hydroelectric power station has an efficiency of 70%. Its output power is 165 MW. Water passes through the turbine of the power station at a rate of 400 000 kg s^{-1}. Find the difference in water levels of the reservoirs in the hydroelectric power station.

pumped hydroelectric storage system 抽水蓄能系統

☆ HKDSE 2016 Paper 2 Q.3

→ This video demonstrates how this question is solved.

Video

DSE practice 2 | Pumped hydroelectric storage system

A pumped hydroelectric storage system has 6 pipelines to deliver water between the upper and the lower reservoirs (Fig a). The upper reservoir can at most contain 9.21×10^{11} kg of water. The water level of the upper reservoir is 150 m above that of the lower reservoir. The system has efficiencies of 80% and 70% in pumping water to the upper reservoir and generating electricity respectively.

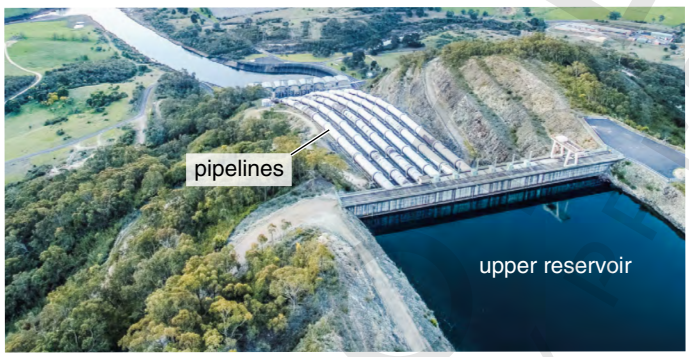


Fig a

What is the energy required to pump water to fill up the upper reservoir from empty? What is the electrical energy generated when all the water flows back to the lower reservoir?

	Energy for pumping water / $\times 10^{15}$ J	**Electrical energy generated / $\times 10^{15}$ J**
A	1.08	0.95
B	1.08	0.76
C	1.69	0.95
D	1.69	1.94

Solution

Total gravitational PE stored

$= mgh = (9.21 \times 10^{11}) \times 9.81 \times 150 = 1.355 \times 10^{15}$ J

Consider the process of pumping water to the upper reservoir.

$$\text{Efficiency} = \frac{\text{energy output}}{\text{energy input}}$$

$$\Rightarrow \text{energy input} = \frac{\text{power output}}{\text{efficiency}} = \frac{1.355 \times 10^{15}}{80\%} = 1.69 \times 10^{15} \text{ J}$$

$\therefore$ The energy for pumping water is 1.69×10^{15} J.

Consider the process of water flowing back to the lower reservoir.

$$\text{Efficiency} = \frac{\text{energy output}}{\text{energy input}}$$

$\Rightarrow$ energy output = energy input $\times$ efficiency

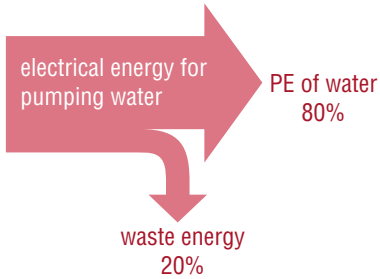
$= 1.355 \times 10^{15} \times 70\% = 9.49 \times 10^{14}$ J

$\therefore$ The electrical energy generated is 9.49×10^{14} J.

$\therefore$ The answer is C.

Common mistake ⚠

Some students may mix up the input energy with the output energy. Note that in the pumping process,



▶ Revision 4 Q27 (p.181)

C Solar energy

Solar energy is the energy from the Sun that is converted into internal energy or electrical energy. Before we study its conversion, let us learn how to measure the energy from the Sun.

DSE goal

Define solar constant.

Solve problems involving the solar constant.

DSE exam

12(2)Q.3, 13(2)Q3.3, 15(2)Q3.3, 17(2)Q3.5, 20(2)Q3.2, 21(2)Q3.3

i Solar constant

The Sun radiates a tremendous amount of energy in all directions. To find out how much energy is radiated to the Earth, scientists measure a value called the **solar constant**.

The solar constant is the total electromagnetic radiation energy radiated at normal incidence by the Sun per unit time per unit area at the mean distance between the Earth and the Sun, measured outside the Earth's atmosphere.

The actual power per unit area fluctuates from 1321 W m^{-2} to 1412 W m^{-2} in a year, due to the varying distance of the Earth from the Sun.

The solar constant is measured in W m^{-2} and its value is about 1366 W m^{-2} as measured by satellites.

The actual value is highly dependent on the weather and can become very small on a cloudy day.

However, if we measure the power of solar radiation per unit area on the Earth's surface on a clear day, the value is smaller than the solar constant and it is approximately equal to 1000 W m^{-2}. This is because the sunlight is partly absorbed by the atmosphere (Fig 4.1p).

For any place on the Earth's surface, the power of solar radiation received is not constant since the Earth is revolving.

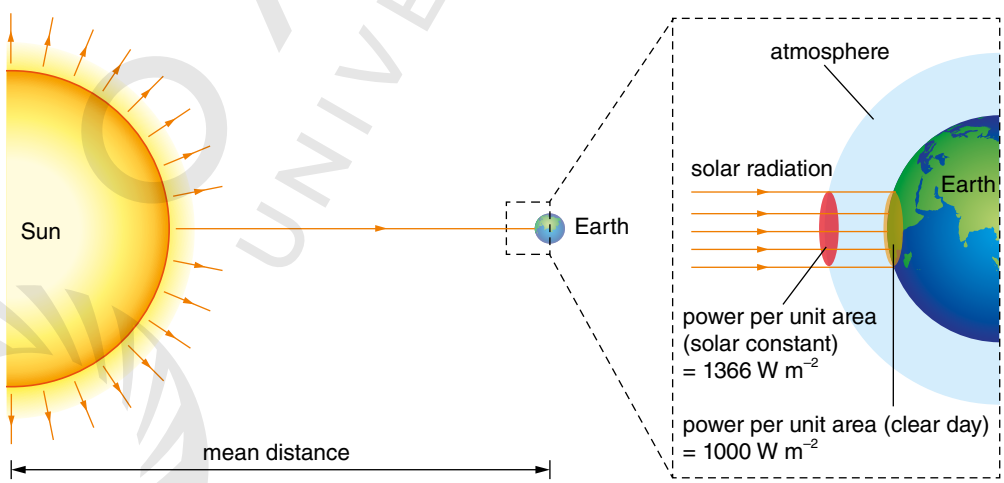


Fig 4.1p *Power of solar radiation measured outside the Earth's atmosphere and on the Earth's surface.*

If sunlight falls on a surface outside the atmosphere at an angle,

Total power received per unit area
= solar constant × cos θ

where θ is the angle between sunlight and the normal of the surface (Fig 4.1q).

Cross-link

The calculation is similar to that of illuminance. See Ch 2.2 p.60 for details.

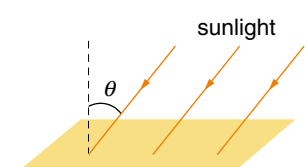


Fig 4.1q *Sunlight making an angle θ with the normal of a surface.*

solar constant 太陽常數

Example 6 | Power received by a surface

A satellite is on the outer surface of the Earth's atmosphere. It has a plate of area 0.64 m^2. Sunlight falls on the plate and the solar constant is 1366 W m^{-2}. Estimate the total power received by the plate if the sunlight is incident

(a) normally,

(b) at 60° (Fig a).

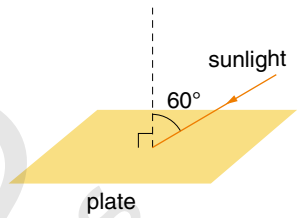


Fig a

Solution

(a) Total power received = solar constant × area
= 1366 × 0.64
= 874 W

(b) Total power received = (solar constant × cos θ) × area
= (1366 × cos 60°) × 0.64
= 437 W

Google Forms Microsoft Forms

Quick check 7

→ E-version in OUP Exercise Platform is also available.

True or false:

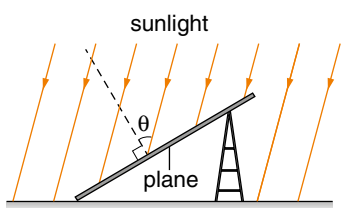
1 The solar constant is the total solar energy received by a unit surface area. (T/F)

2 The solar constant is measured outside the Earth's atmosphere. (T/F)

3 The power of solar radiation per unit area measured on the ground is always higher than the solar constant. (T/F)

Learn by practice 6

1 Assume that 1050 W m^{-2} of solar power reaches the Earth's surface. A tilted plane of area 2 m^2 on the Earth's surface is illuminated by sunlight. The total power received by the plane is 1800 W. Find the angle θ between sunlight and the normal of the plane.

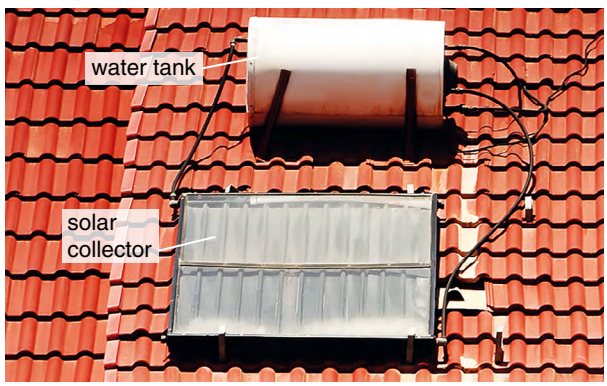


ii Solar thermal panels

Cross-link
For heat transfer by radiation, refer to Book 1 Ch 2.2.

Solar energy can directly be used for heating, for example, solar water heaters (Fig 4.1r) and solar cookers (Fig 4.1s). The solar collectors can be in different designs. They may be flat or curved in shape and dull black or silvery in colour.

(i)



(ii)

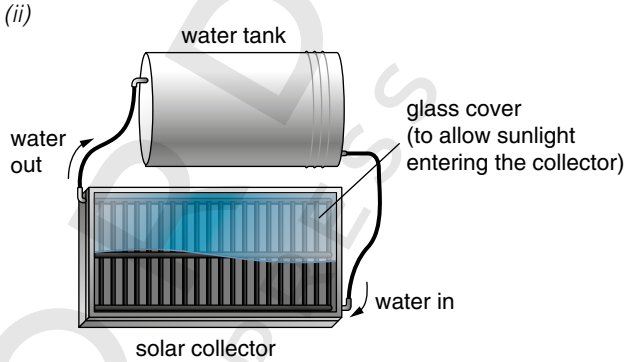


Fig 4.1r *(i) A solar water heater and (ii) its structure.*

(i)



(ii)

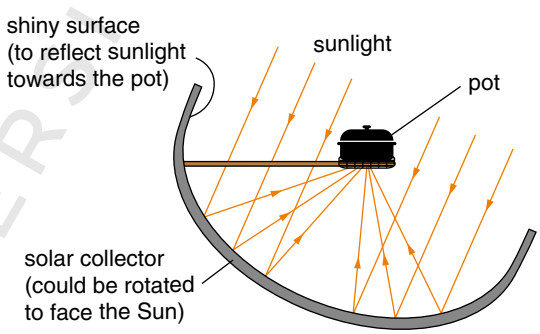


Fig 4.1s *(i) A solar cooker and (ii) its structure.*

Example 7 Solar water heater

A solar water heater has a flat collector of area 4 m^2. It is illuminated normally by sunlight for 30 minutes to heat water. The solar constant is 1366 W m^{-2}. The atmosphere absorbs 25% of the power of solar radiation. If the efficiency of the heater is 45%, estimate the solar energy absorbed by the water.

Solution

Power of solar radiation per unit area on the Earth's surface
$= 1366 \times (1 - 25\%) = 1025$ W m^{-2}

Energy collected by the collector $= 1025 \times 4 \times (30 \times 60) = 7.38 \times 10^6$ J

Energy absorbed by the water $= 7.38 \times 10^6 \times 45\% = 3.32 \times 10^6$ J

Know more

Concentrated solar power

A *concentrated solar power* (CSP) plant has a similar working principle as a solar cooker, but the solar heat is used further to generate electricity. A large array of movable mirrors concentrates sunlight to the top of the tower and heat the *heat-transfer fluid* there up to higher than 400 °C. The hot fluid is then circulated to the steam generator to boil water. Steam is produced to drive a turbine, which in turn drives a generator to produce electricity.

A CSP plant requires vast land to install the mirrors. For example, the Gemasolar power plant as shown below has 2650 movable mirrors that placed across an area of about 360 football fields. Its maximum power output is 20 MW, which is enough to supply power to 27 500 homes.



Gemasolar power plant in Spain.

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

Quick check 8

1 Which of the following best describes the energy conversion in solar heating?

A light energy → internal energy

B light energy → electrical energy

C light energy → chemical energy

D light energy → potential energy

Learn by practice 7

1 A solar water heater has a flat collector of area 2 m^2. When sunlight is incident on the collector normally, the power received per unit area is 1000 W m^{-2} and its total output power is 250 W. What is the efficiency of the heater?

concentrated solar power 太陽能聚熱發電　　heat-transfer fluid 熱媒油

DSE goal

Describe how a solar cell works.

DSE exam

15(2)Q3.2, 16(2)Q3.4, 22(2)Q.3, 23(2)Q3.6

In 1954, researchers used semiconducting materials (silicon) to make the first practical solar cell.

Vanguard 1, which was launched in 1958, was the first artificial satellite with solar cells installed.

iii Solar cells

A **solar cell**, or **photovoltaic cell**, is a device that converts light energy into electrical energy. Solar cells are usually assembled into **solar panels**. The electricity produced may be stored in batteries for future use. From small devices (Fig 4.1t) to large scale power stations (Fig 4.1u), the size of solar panels can vary greatly.



(i) Solar powered calculator.



(ii) Solar powered garden lamp.

Fig 4.1t Solar panels in small devices.



Fig 4.1u Large solar panels in a solar power station.

Large scale solar farms, consisting of over 10 000 solar panels, are usually located in remote areas (Fig 4.1v). Some of them have power output higher than 500 MW which is comparable to a conventional fossil fuel power station.



Fig 4.1v A large scale solar farm in the desert.

solar cell 太陽能電池　　photovoltaic cell 光伏電池　　solar panel 太陽能電池板

Cross-link

See p.52–53 in Ch 2.1 to review the properties of p-type and n-type semiconductors.

The structure of a solar cell is similar to that of an LED. However, a solar cell is a power source while an LED is a normal circuit component.

Basically, a solar cell consists of a layer of n-type semiconductor attached to a layer of p-type semiconductor. This structure creates an electric field pointing from the n-side to the p-side at the p-n junction. When exposed to sunlight, an e.m.f. is set up across the two layers (Fig 4.1w). This is called the *photovoltaic effect*. A direct current is driven if it is in a complete circuit.

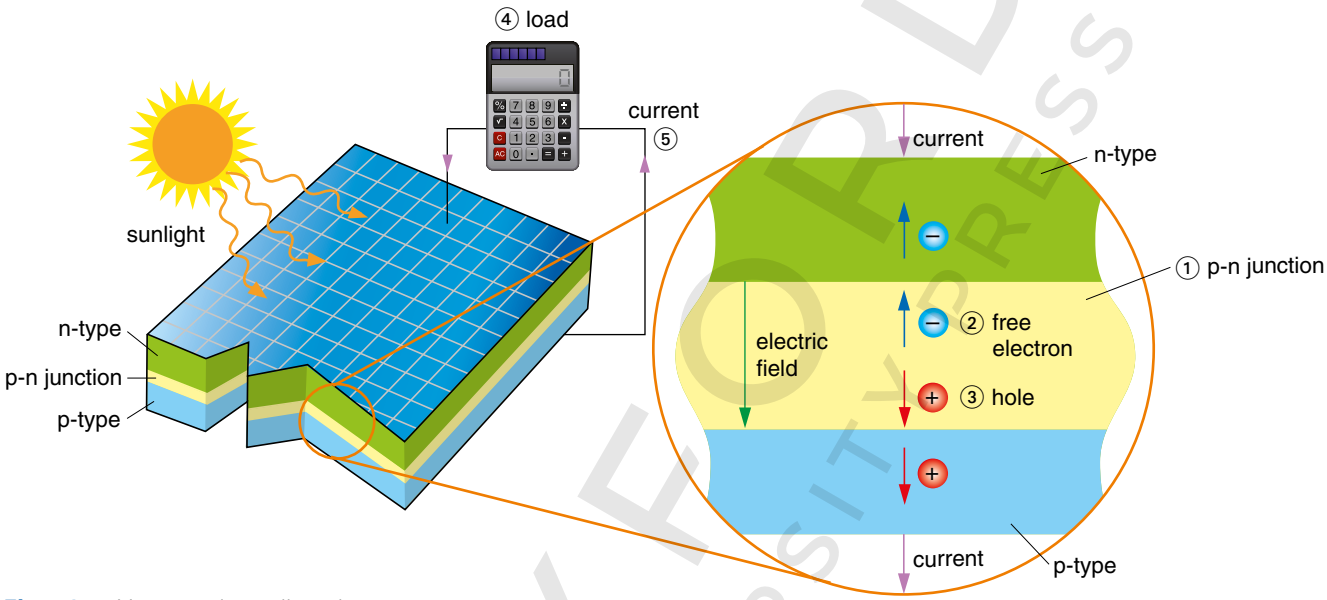


Fig 4.1w *How a solar cell works.*

In a solar cell, light energy is absorbed to form electron-hole pairs. In an LED, light energy is released as a result of electron-hole combination.

① At the p-n junction, electrons in atoms absorb sunlight and become free electrons. This leaves holes behind.

② The free electrons move against the direction of the electric field towards the n-side and migrate to the top surface of the cell.

Recall that holes are positive charge carriers.

③ The holes move along the direction of the electric field towards the p-side and migrate to the bottom surface of the cell.

④ If the cell is connected to a load, e.g. a calculator, free electrons will flow from the n-side to the p-side via the load, and rejoin the holes at the p-side.

⑤ As a result, a d.c. current flows from the p-side to the n-side via the load.

Under higher intensity of light, more free electrons move in a solar cell and hence a larger current is driven. However, the output voltage of the solar cell, which depends on the material of the p–n junction, remains fairly constant.

The efficiency of a solar panel is lower at a high temperature. A cooling system may be needed to keep the panel within certain temperatures.

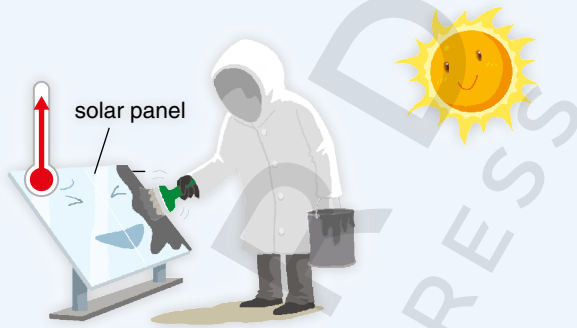
A typical solar cell only has an efficiency of about 10–20%. A transparent anti-reflection film is usually added to the top of a solar panel to improve its absorption of light. This makes the panel look dark in colour.

photovoltaic effect 光伏效應

Right and wrong

✗ Painting the solar panel in dark colours could produce more electrical energy.

✓ Painting the solar panel in dark colours will convert more solar energy into internal energy but not electrical energy.



More and more buildings have installed photovoltaic (PV) systems on their rooftops (Fig 4.1x). The system is mainly composed of one or more solar panels and an inverter (Fig 4.1y). The inverter converts the output d.c. of the PV system into a.c., so that the electricity produced can be fed into the power grid.



Fig 4.1x *A photovoltaic system installed on a rooftop.*

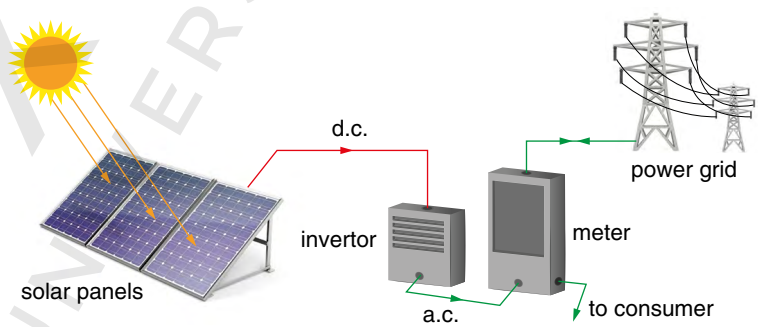


Fig 4.1y *Schematic diagram of the photovoltaic system.*

Solar panels need little maintenance since there is no moving part. Only regular cleaning is needed.

Dust or dirt on the surface of the solar panels will reduce the amount of electricity generated. Therefore, simple cleaning work should be performed on a regular basis.

Know more

Feed-in Tariff

A Feed-in Tariff (FiT) is a payment for electricity fed into the supply grid from a renewable energy source. In Hong Kong, only photovoltaic systems and wind power systems are eligible for participation in FiT Scheme at present. To promote renewable energy among the public, the two electricity companies offer a rate higher than the normal electricity tariff rate to purchase all the units of electricity generated with renewable energy source by their registered customers.

The automatic weather stations in remote areas or islands have no supply of main electricity.

- Solar cells can be used in remote and isolated locations (Fig 4.1z).

Solar power is available only during daytime and the output power varies with the weather. To ensure continuous power supply, a rechargeable battery is often used to store some of the electricity generated as reserve.

Electrical energy is stored as chemical energy in the battery.

See Table 4.1a on p.162 for the comparison of different energy sources.

- Wind power is also common to use to supplement solar power.

The world's solar power generation grew by over 25% each year in the past decade. As the price of solar panels continue to decline, solar power generation is expected to maintain strong growth in the near future.



Fig 4.1z *A remote automatic weather station powered by a solar cell.*

Google Forms Microsoft Forms

Quick check 9

→ E-version in OUP Exercise Platform is also available.

1 Determine whether the following statements about the generation of electricity in a solar panel are true.

(a) Electrons move to the n-side and holes move to the p-side due to the electric field developed at the p-n junction. (T/F)

(b) Electrons flow from the p-side via the external circuit to the n-side. (T/F)

(c) In the p-n junction, electrons in atoms absorb sunlight and become free electrons, leaving holes behind. (T/F)

2 A solar panel converts only about 11% of solar energy falling on it into the energy stored in the battery. Determine whether the following statements are true.

(a) Energy conversion during the charging process is: solar energy → electrical energy → chemical energy (T/F)

(b) The efficiency of the solar panel in charging the battery is 11%. (T/F)

(c) The solar panel has a low efficiency because sun rays are not always normal to the panel. (T/F)

(d) The solar panel has a low efficiency because there is loss of solar radiation due to absorption by the atmosphere. (T/F)

(e) The solar panel has a low efficiency when it is cloudy. (T/F)

(f) The output voltage remains constant when the incident light intensity increases. (T/F)

Now we can compare the energy sources for electricity generation we have learned (Table 4.1a).

Table 4.1a *Comparison of some energy sources.*

	Fossil fuel	Nuclear power	Wind power	Hydroelectric power	Solar power
Energy conversion	chemical energy in fuel → internal energy of water → KE of turbine → electrical energy	nuclear energy in nuclei → KE of fission products → internal energy of water → KE of turbine → electrical energy	KE of air → KE of rotor → electrical energy	PE of water → KE of water → KE of turbine → electrical energy	light energy from the sun → electrical energy
Advantages	• stable output		• inexhaustible • cause less pollution		
	• can be used at small scale	• does not cause air pollution during the production of electricity	• can be used at small scale	• stable output • dam can control floods	• available everywhere • can be used at small scale
Disadvantages	• exhaustible • cause pollution in mining • cause thermal pollution as hot water is released after being used as coolant		• unstable output • noisy • high transmission cost (wind farms usually far from cities) • needs stable and strong wind over a long period of time	• land is flooded by reservoir • greatly affects natural river environment • needs large rivers	• available only in daytime • unstable output • needs stable and strong sunlight over a long period of time
	• causes air pollution and global warming	• poses threat to people and environment if accident happens • produces radioactive waste • risk of nuclear proliferation			

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Know more

Space-based solar power NSE

Solar energy is one of the fastest-growing sustainable energy sources. The use of solar panel, however, is not fully utilized on Earth. Without the interception of the atmosphere, solar panels can collect much more solar energy in space than on Earth. The solar energy collected in space could be transmitted to the ground in the form of microwave. The ground facilities receiving the microwave will then convert the radiation into electrical energy.

Moreover, space-based solar power stations can operate for 24 hours each day. Therefore, they attract great interest of governments around the globe for its potential in solving the energy crisis.

Energy source is one of the major topics in resource security. In order to safeguard energy security and boost the energy reserve capacity, our country plans to build the first ever space-based solar power station and launch it by 2028.



The solar cells in space is illuminated by the Sun nonstop.

STSE

Development of renewable energy in the world

The use of renewable energy is growing in many countries. In fact, different kinds of renewable energy are suitable for different geographical regions. Can you explain why?

Although non-renewable energy sources may be exhausted in the future, they are unlikely to be completely replaced by renewable energy sources shortly as the renewable energy sources have many limitations. Can these limitations be overcome?

Useful link

Exercise 4.1

Take 1 u = 1.661×10^{-27} kg = 931 MeV, $e = 1.60 \times 10^{-19}$ C.

1 The binding energies of four nuclides are shown below. Which one is the most stable?

	Nuclide	Binding energy / MeV
A	Li-5	26.3
B	Mg-29	235
C	Fe-48	385
D	Pu-243	1830

***2** Which of the following statements about a solar cell is/are correct?

(1) It generates a.c.

(2) There is an electric field developed within the junction between the n-type and the p-type layers.

(3) It emits light when it is connected in a closed circuit.

A (1) only

B (2) only

C (1) and (2) only

D (1) and (3) only

★**3** Which of the following statements about the moderator in a nuclear reactor is/are correct?

(1) It lowers the KE of the neutrons produced by fission.

(2) It lowers the rate of fission.

(3) It absorbs the neutrons produced by fission.

A (1) only

B (3) only

C (1) and (2) only

D (2) and (3) only

★**4** Two wind turbines X and Y are located in the same place and have the same efficiency. The length of the blades of X is L while that of Y is $2L$. A wind blows normal to the two turbines at the same speed and the power output of X is P. Find the power output of Y.

A P

B $2P$

C $4P$

D $8P$

★**5** A pumped hydroelectric storage system uses 8.6×10^{13} J of energy to pump 3.30×10^{10} kg of water to the upper reservoir, which is at a vertical distance of 200 m above the lower reservoir. When the same amount of water flows back to the lower reservoir, 4.4×10^{13} J of electrical energy is generated. What are the efficiencies of the system in pumping water and generating electricity respectively?

	Efficiency in pumping water	Efficiency in generating electricity
A	75%	68%
B	68%	75%
C	75%	51%
D	51%	68%

★**6** (a) There are 1000 kg of fuel in a nuclear power station. In a certain period of time, about 0.1% of the fuel's mass is converted to energy in the reactor and is further converted into electrical energy with an efficiency of 40%. Estimate the electrical energy output of the power station within this period.

(b) The schematic diagram of a nuclear reactor is as shown (Fig a). Shade the part(s) in which the water will be in contact with radioactive substances under normal operation.

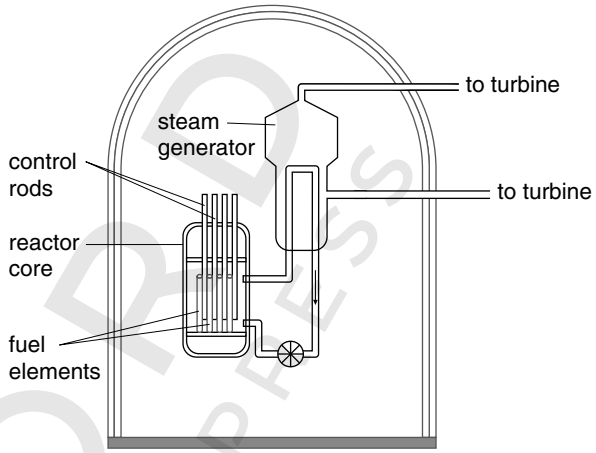


Fig a

★**7** Energy is released in the nuclear reaction shown.

$$^{7}_{3}\text{Li} + {}^{1}_{1}\text{H} \rightarrow 2\,^{4}_{2}\text{He}$$

Explain whether the total binding energy of the reactants or that of the product is higher.

★**8** Solar panels of total area 15 m^2 are placed on the roof of a house. The panels are inclined at 30° to the roof floor (Fig b). Assume that 1000 W m^{-2} of solar power reaches the roof floor vertically.

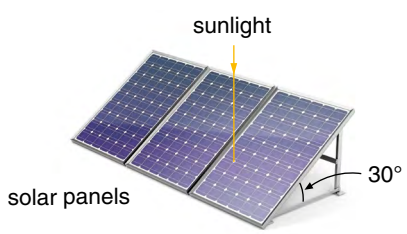


Fig b

(a) What is the total power received by the solar panels?

(b) The conversion efficiency of the solar panels is 12%. Estimate the electrical power output of the solar panels.

★**9** A solar water heater has a flat collector of area 6 m^2. The efficiency of the heater is 55%. If the power of solar radiation received per unit area by the collector is 1000 W m^{-2}, estimate the time needed to raise the temperature of 8 kg of water for 5 °C. The specific heat capacity of water is 4200 J kg^{-1} $°C^{-1}$.

★**10** Plutonium-239 (Pu-239) can be used as the fuel in nuclear reactors. The following equation shows a nuclear reaction that may take place.

$$^{239}_{94}\text{Pu} + {}^{1}_{0}\text{n} \rightarrow {}^{90}_{A}\text{Sr} + {}^{B}_{56}\text{Ba} + 2{}^{1}_{0}\text{n}$$

Given: binding energy of $^{239}_{94}$Pu = 1810 MeV

binding energy of $^{90}_{A}$Sr = 783 MeV

binding energy of $^{B}_{56}$Ba = 1210 MeV

(a) Find the energy released in this reaction.

(b) Which nuclide, $^{239}_{94}$Pu, $^{90}_{A}$Sr, or $^{B}_{56}$Ba, is the most stable?

★**11** The maximum power output of a wind turbine is 0.9 MW when the wind blows at 13 m s^{-1}. The density of air is 1.23 kg m^{-3} and the efficiency of the turbine is 30%.

(a) Describe the energy conversion in a wind turbine in generating electricity.

(b) Estimate the length of the rotor blades.

(c) Explain why the turbines in a wind farm should not be built close to each other.

★**12** Figure c shows a graph of binding energy per nucleon for some nuclides against their mass number.

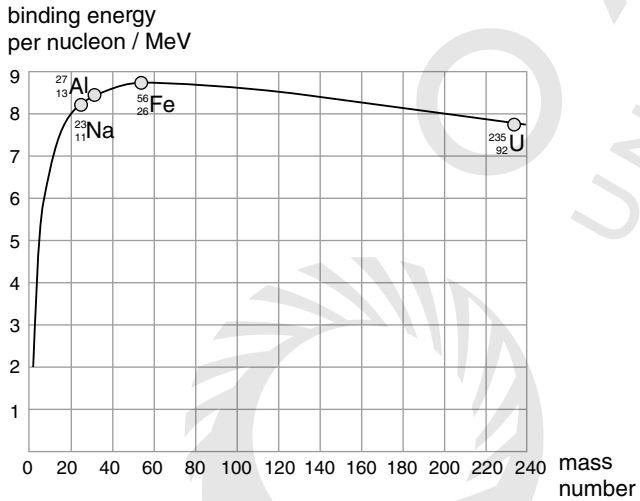


Fig c

Which of the nuclides shown in the diagram

(a) is the most unstable?

(b) need the largest amount of energy to split itself into individual nucleons?

(c) will **not** release energy when splitting into smaller nuclei?

★**13** In a hydroelectric power station, the difference in water levels between the reservoirs is 150 m (Fig d). The output power of the station is 5 GW and its efficiency is 75%.

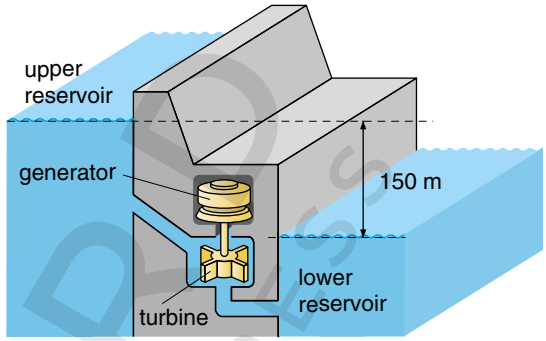


Fig d

(a) Describe the energy conversion in a hydroelectric power station in generating electricity.

(b) Find the flow rate (in kg s^{-1}) of water through the turbine.

★**14** The power of solar radiation per unit area at normal incidence on the atmosphere of Mars is 590 W m^{-2}. Scientists plan to build a solar power station on Mars (Fig e). Suppose that the atmosphere of Mars absorbs 12% of the solar power. The solar panels in the project should have a total output power of 6000 W when sunlight falls normally on them.



Fig e

(a) If the efficiency of the solar panels is 20%, what is the total area of panels required?

(b) What is the total power output of the panels when sunlight makes an angle of 40° to the normal of each panel?

4.2 Environmental impact of energy consumption

Flipped classroom
Watch a video and answer the questions.

DSE goal
Discuss the impact of extraction, conversion, distribution and use of energy on the environment and society.

As an introduction, ask Ss what the cost of using energy is. 'Cost' not only refers to the amount of money we pay, but also all the things we sacrifice (e.g. natural landscape, fresh air).

1 Impact on environment and society

All types of energy sources undergo some processes before the energy they provide becomes 'usable'. The following are four common processes (Fig 4.2a).

1 **Extraction:** taking the energy source from the environment



2 **Conversion:** converting the energy source into the desired forms



3 **Distribution:** distributing the energy (e.g. electricity) to end users



4 **Consumption:** using the energy



Fig 4.2a *How the energy from a source becomes 'usable'.*

We will take fossil fuels as an example and discuss the impact of these processes on the environment and society.

Extraction

Fossil fuel mining may lead to:

- Deforestation: This can cause erosion and landslides and destroy habitat for animals and plants.



- Release of toxic chemicals and heavy metals: This contaminates the soil, rivers and groundwater.

For example, mining usually results in exposed areas with coal heaps and significant sulphur content. This leads to highly acidic drainage when it rains. The drainage can cause severe pollution.



Conversion

Sulphur dioxide and nitrogen oxides are transported by wind, sometimes over long distances. They slowly transform to pollutants such as nitric acid and sulphuric acid droplets.

Consequences of burning fossil fuels include:

- Release of greenhouse gases into the atmosphere, causing global warming.



- Release of pollutants such as sulphur dioxide and nitrogen oxides, leading to air pollution and the production of smog and acid rain.
- Discharge of hot water (after used as coolant) into rivers and seas, causing *thermal pollution* and endangering marine life.





thermal pollution 熱污染

Distribution

Overhead and underground power cables are used to transmit electricity at high voltages. They produce strong electromagnetic fields which may:

- Increase the chances of getting blood cancer.
- Restrict land use in the area.



Consumption

All the energy we use will finally change into internal energy. Using more energy may lead to:

- Faster consumption of fossil fuels, which may be depleted in the future.
- Heat island effect, i.e. a significantly higher temperature in urban areas.

In areas with tall buildings, heat and pollutants form a haze dome that prevents warm air from rising and being cooled at a normal rate, especially in the absence of strong winds. This results in heat island effect.



Know more

West-to-east power transmission program NSE

Resource security is of vital importance in national strategy and national development. Our country has implemented the *west-to-east power transmission program* to balance the electricity supply and demand in different regions. It transmits the electricity surplus in the western regions which are rich in power-generating resources to the eastern regions which have great demand for electricity to boost economic activities.

(Source: Our country's Tenth Five-Year-Plan for National Economic and Social Development)



west-to-east power transmission program 西電東送計劃

DSE goal
Discuss effect of greenhouse gases on global warming.

2 Effect of greenhouse gases on global warming

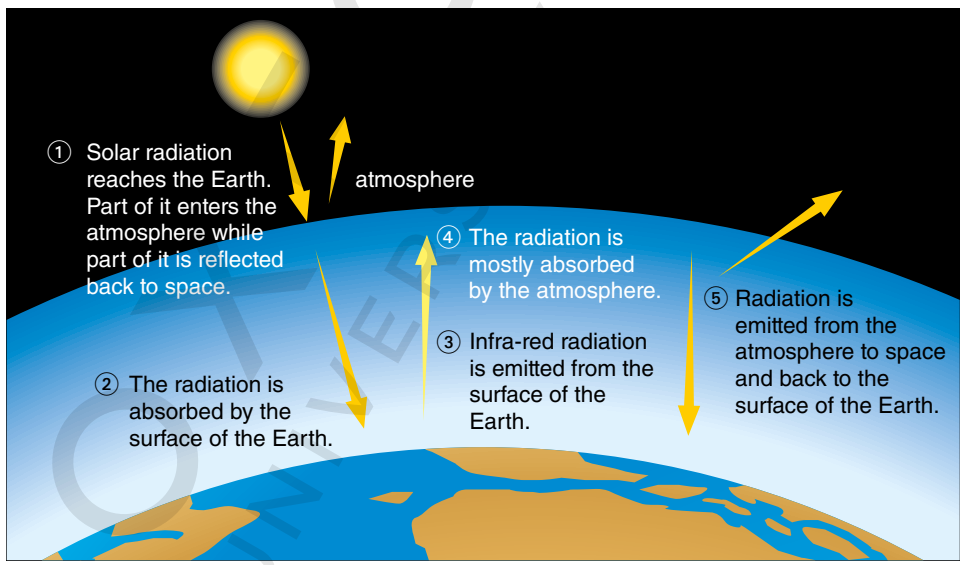
Video

→ This video shows the greenhouse effect and the problems caused by global warming.

a Greenhouse effect

The **greenhouse effect** is essential for life on Earth. **Greenhouse gases**, including water vapour, carbon dioxide (CO_2), methane (CH_4), etc., exist naturally in our atmosphere. These gases absorb and emit infra-red radiation, producing the greenhouse effect. With this effect, the average surface temperature of the Earth can be maintained at about 15 °C.

Figure 4.2b explains what the greenhouse effect is. If the Earth did not have the atmosphere, the heat emitted by the Earth would simply pass outwards into space and the Earth would have an average temperature of only about −18 °C!



The Earth's surface is much colder than the Sun, so the radiation it emits has longer wavelengths than solar radiation. Radiation with longer wavelengths is more easily absorbed by the atmosphere.

Fig 4.2b *Greenhouse effect.*

The greenhouse effect is a natural process by which the atmosphere of the Earth traps some of the thermal radiation from the Earth's surface and warms the Earth.

b Global warming

Ts may emphasize that greenhouse effect is essential for life, but enhanced greenhouse effect causes global warming which poses a serious threat to life on Earth.

The amount of greenhouse gases in the atmosphere has been increasing rapidly over the last few decades (Fig 4.2c on p.168). This has enhanced the greenhouse effect. Since the late 19th century, the average surface temperature of the Earth has increased by 0.7 °C. This climate change is called **global warming**.

greenhouse effect 温室效應　greenhouse gas 温室氣體　global warming 全球暖化

✎ One part per million (ppm) denotes one part per 1 000 000 parts. The concentration of CO_2 in the atmosphere is currently about 400 ppm, which means there are 400 CO_2 molecules in 1 000 000 air molecules.

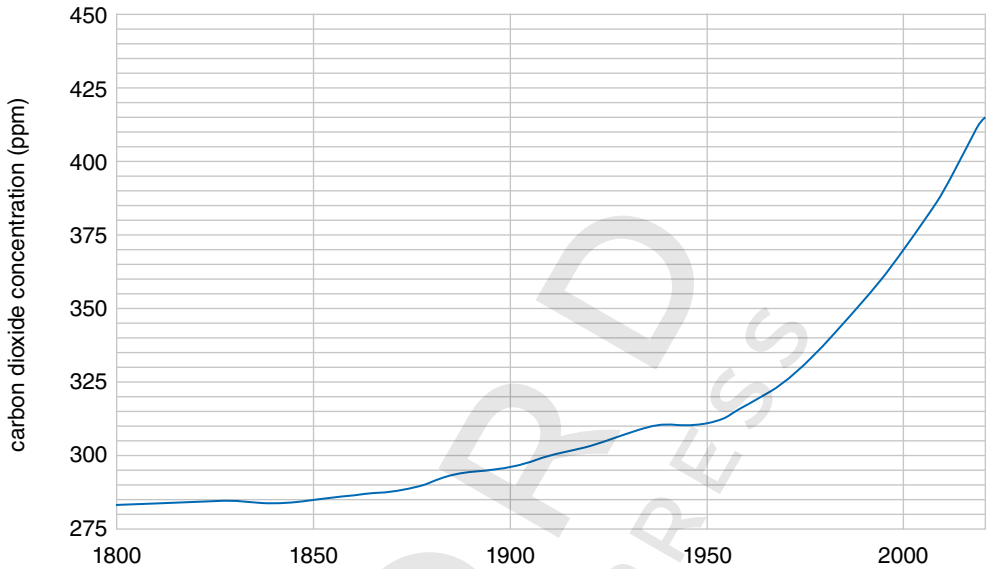
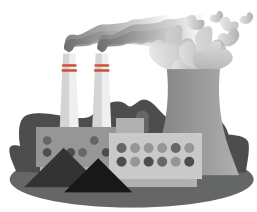


Fig 4.2c *Graph showing the amount of CO_2 in the atmosphere starting from 1800. (Data obtained from NOAA website in Aug 2022)*

Why is the amount of greenhouse gases increasing so fast? The main causes are all related to humans as follows.

What can we do to minimize the emission of greenhouse gases? ▶

1 Burning fossil fuels



This releases a large amount of greenhouse gases including carbon dioxide and methane.

✎ Forests are cleared to create new pastures for livestock and grow crops such as soya beans to feed them. It is a major driver of deforestation.



2 Destruction of forests

Plants absorb carbon dioxide. However, some forests are burnt and cleared for development. This leads to an increase in the amount of carbon dioxide in the atmosphere.

3 Population growth

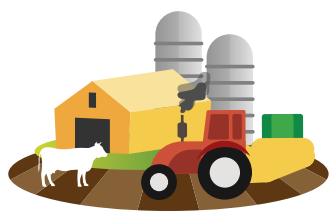
The increasing global population leads to a higher demand for energy and food. As a result, more greenhouse gases are released from burning more fossil fuels, producing more livestock and crop, and wasting more food.

It is estimated that nearly one-third of all food produced for human consumption is lost or wasted! ▶



Animals like cattle and sheep give out methane due to *enteric fermentation*. Agriculture is the largest human-influenced source of methane.

The production, transportation, and handling of food generate significant carbon dioxide emissions. The wasted food ends up rotting in landfills and releasing methane.



enteric fermentation 腸內發酵

Research shows that global warming leads to the following:

1 **Rise in sea levels**
As the Earth becomes warmer, the polar ice sheets melt. This causes the sea levels to rise and thus more coastal flooding.

2 **Changes in climate**
Extreme weather conditions, e.g. heatwave, rainstorms or droughts, becomes more severe or more likely to occur. Also, tropical cyclones may become more intense.

3 **Spread of diseases**
A warmer climate favours the spread of diseases like *dengue fever*.

Dengue fever is spread through mosquito bites, and a warmer climate facilitates the reproduction of mosquitoes.

Warmer air holds more moisture than cooler air. Therefore, as global temperatures rise, the total amount of water vapour in the atmosphere also increases, further amplifying the warming effect. This human-induced climate change has already affected all regions on Earth. So, we have to tackle the climate crisis decisively and implement changes accordingly.

Google Forms Microsoft Forms

Quick check 10

→ E-version in OUP Exercise Platform is also available.

True or false:

1 The greenhouse effect has only negative effect on us. (T/F)
2 Greenhouse gases are only released in human related activities. (T/F)
3 Global warming maintains our planet at a suitable temperature for life. (T/F)

Know more

Resource security NSE

Energy source is important in social and economic development, so we must ensure a sustainable and stable supply of energy. Electricity, town gas, liquefied petroleum gas and fuel oil are the most common energy sources in Hong Kong. Most of them are directly imported or converted from imported fuels. We can help reduce the energy resource consumption in our daily lives to support resource security.

Ecological security NSE

When we consume energy, for example, by burning fossil fuels, greenhouse gases are released. The gases will enhance global warming, which in turn will endanger the natural world. We should help safeguard ecological security and minimize damage to nature.

One of the methods is to reduce waste and carbon emissions. The HKSAR Government has implemented various measures to help reduce waste and carbon emissions, and planned to set out more proactive strategies and measures to strive to achieve carbon neutrality.

Try to get more information about these measures on the Internet.

Useful link

(Source: 'Twenty Major Fields' in National Security Education Day website)

dengue fever 登革熱

DSE goal

Analyse the consumption data for different fuel types in Hong Kong and their specific purposes.

3 Energy consumption in Hong Kong

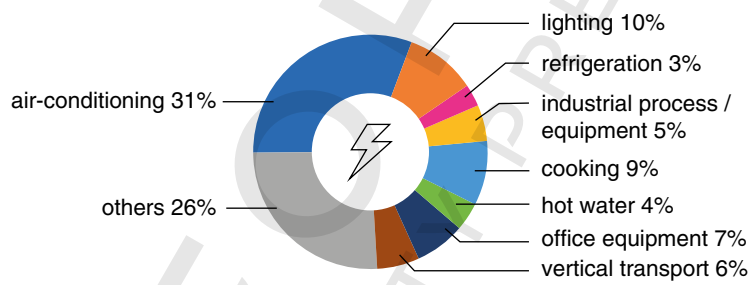
In order to save energy, we need to know how much energy is used for different purposes. Information about energy consumption in Hong Kong can be found in *Hong Kong Energy End-use Data*. You may visit the corresponding website to see it. Let us have a quick look here.

The Electrical and Mechanical Services Department has been publishing the *Hong Kong Energy End-use Data* every year since 1998.

Useful link

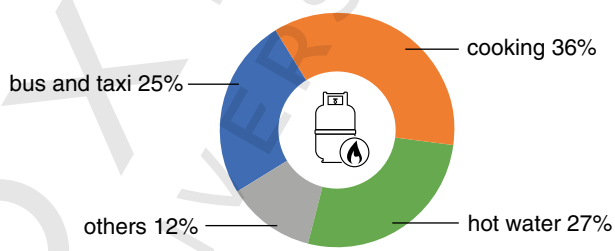
The energy data are classified into three fuel types—'electricity', 'town gas and liquefied petroleum gas (LPG)' and 'oil and coal products'. Figure 4.2d shows how these fuels were used in 2020.

(i) Electricity



Town gas and LPG are refined from oil or coal.

(ii) Town gas and LPG



Oil and coal products include gasoline, diesel, etc.

(iii) Oil and coal products

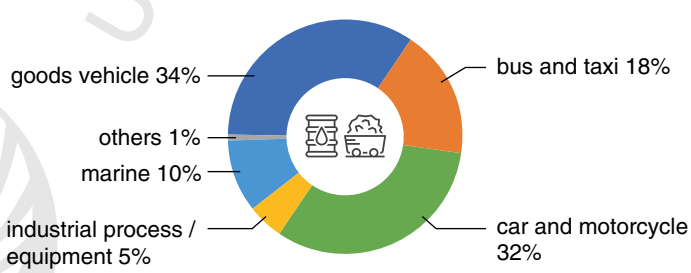


Fig 4.2d *End uses of fuels in Hong Kong in 2020. (Data obtained from EMSD website in Oct 2022)*

The data show that:

1. Electricity was mainly used in air conditioning, lighting and cooking.
2. Town gas and LPG were mainly used in cooking and transport.
3. Oil and coal products were mainly used in transport.

After studying this elective part, you should be able to make informed decisions on issues related to energy production and consumption. You should also consider the energy sustainability so that you will not sacrifice the future generations while using energy.

Exercise 4.2

1 Which of the following contribute to the increasing amount of greenhouse gases in the atmosphere?

(1) Deforestation

(2) Transportation

(3) Agriculture

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

2 Many people are concerned about global warming.

(a) Describe briefly how the use of fossil fuels contributes to this phenomenon.

(b) Describe briefly how global warming affects us.

(c) Suggest two actions that can be taken in daily life to help reduce global warming.

3 (a) Give two examples of greenhouse gases.

(b) Suggest a source of each type of greenhouse gas in (a).

4 Suggest how each of the following energy sources impacts the environment.

(a) Nuclear power

(b) Hydroelectric power

(c) Wind power

***5** Compared to a rural area (Fig a), energy consumption in a modern city like Hong Kong (Fig b) is much higher.



Fig a



Fig b

(a) Suggest two activities that account for a large proportion of the energy consumption in Hong Kong.

(b) Hence, by comparing the environment in a rural area and in Hong Kong, explain how their energy consumption patterns differ.

Key concepts 4

1 Energy sources

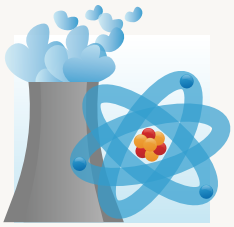
Non-renewable

- finite supply

Fossil fuels



- formed from the organic remains of prehistoric plants and animals
- examples include coal, oil and natural gas
- burned to release energy



Nuclear power

- is related to the binding energy of a nucleus

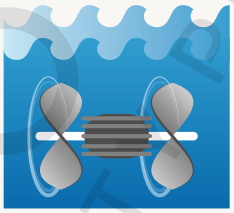
Renewable

- inexhaustible

Wind power



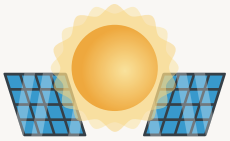
- make use of KE of air



Hydroelectric power

- make use of KE of water

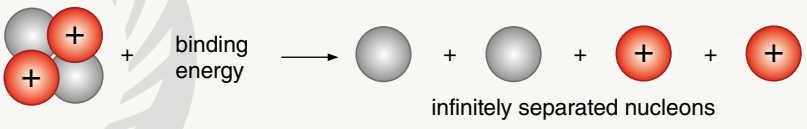
Solar power



- make use of energy from the Sun

2 Nuclear power

Binding energy E_b

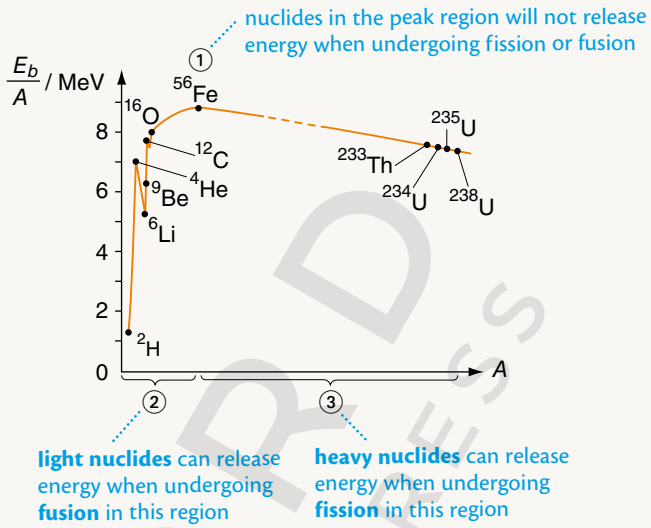


- amount of energy required to completety separate all the constituent nucleons of a nucleus
- E_b (in J) = Δm (in kg) × c^2
- E_b (in Mev) = Δm (in u) × 931

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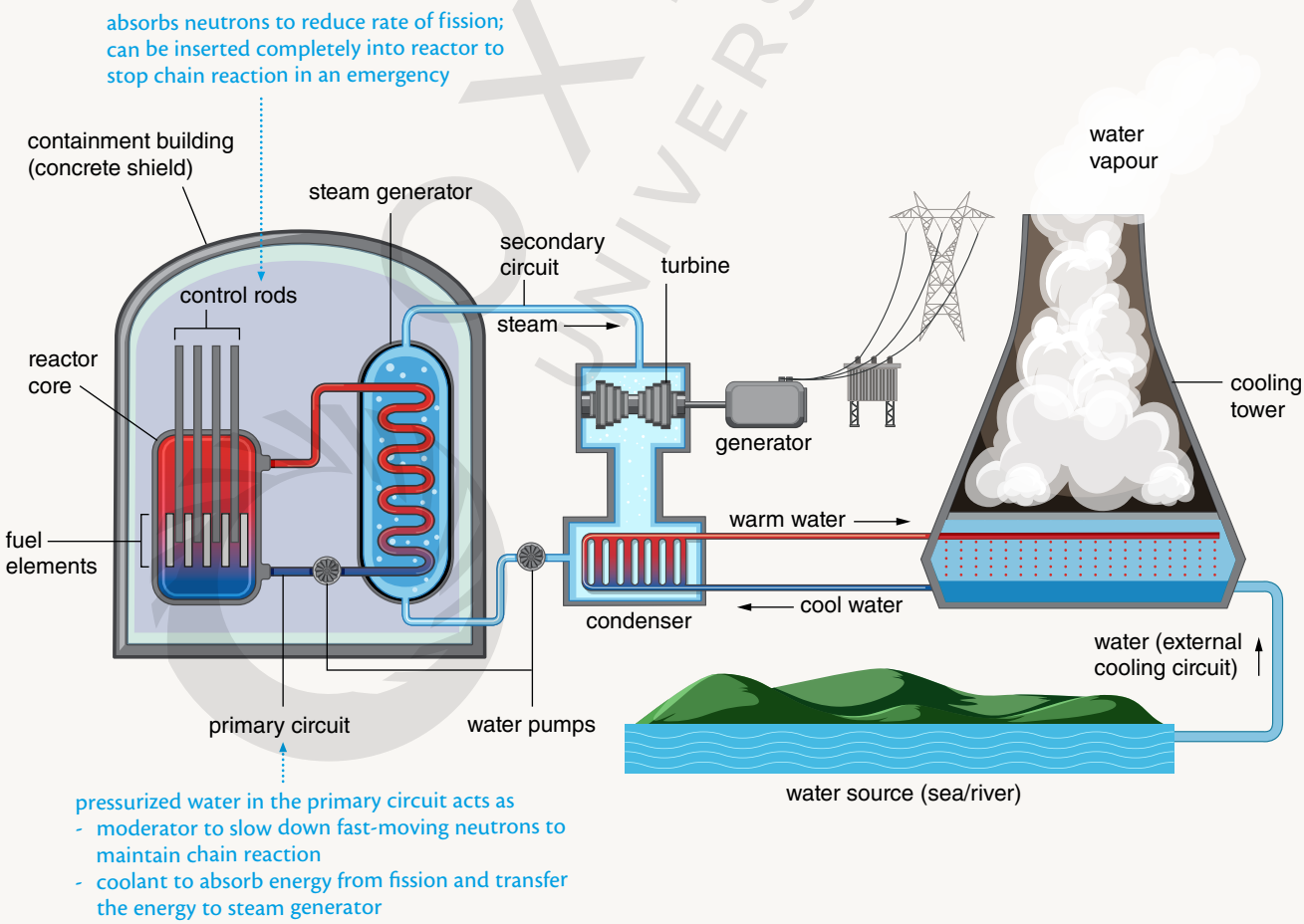
Binding energy per nucleon $\frac{E_b}{A}$

- larger $\frac{E_b}{A} \Rightarrow$ nuclide more stable



Nuclear reactor

- energy from fission → KE of water (steam) molecules → KE of turbine → electrical energy
- structure of pressurized water reactor (PWR):



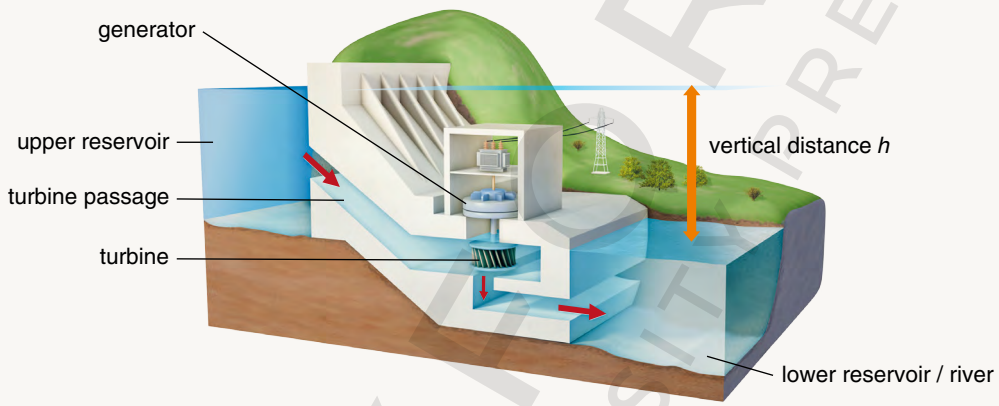
3

Hydroelectric power

- PE of water → KE of water → KE of turbine → electrical energy
- output power P of hydroelectric power station:

power of station (W) $P = \dfrac{\text{loss in PE of water}}{\text{time taken}} \times \text{efficiency} = \dfrac{mgh}{t} \times \eta$

mass of water (kg); **acceleration** due to gravity (m s^{-2}); **distance** between upper and lower reservoirs (m); **time** interval (s); **efficiency** of power station (%)



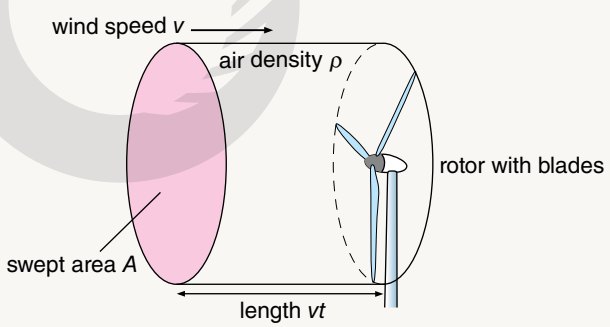
4

Wind power

- KE of air → KE of rotor → electrical energy
- maximum output power P of wind turbine:

maximum **power** of turbine (W) $P = \frac{1}{2}\eta\rho A v^3$

efficiency of turbine (%); **wind speed** (m s^{-1}); **density** of air (kg m^{-3}); sweep **area** of blades (m^2)

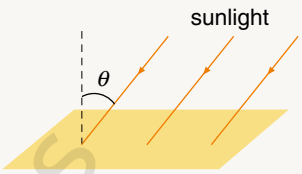


Q27 Exam report: In (a)(i), most candidates mentioned that the turbine will not move when the wind speed is too low, but few of them pointed out that it is due to the friction between the contact surfaces. Some candidates knew that the turbine will be damaged when the wind speed is high, however, not many went on to elaborate that the turbine is actually shut down to prevent such damage. Part (a)(ii) was well answered. In (b)(i), many took it for granted that the efficiency of the wind turbine remained the same at different wind speeds. Instead of finding the required wind speed from the graph provided, they tried to calculate the 'expected' wind speed by using an incorrect efficiency. In (b)(ii)(I), most candidates were able to find the total output power of the wind farm, but some failed to relate it precisely with the time period within which the pumped hydroelectric storage system needs to generate electricity. In (b)(ii)(II), quite a number of them mixed up the pump's input power and output power, and as a result employed an incorrect expression of efficiency in the calculation.

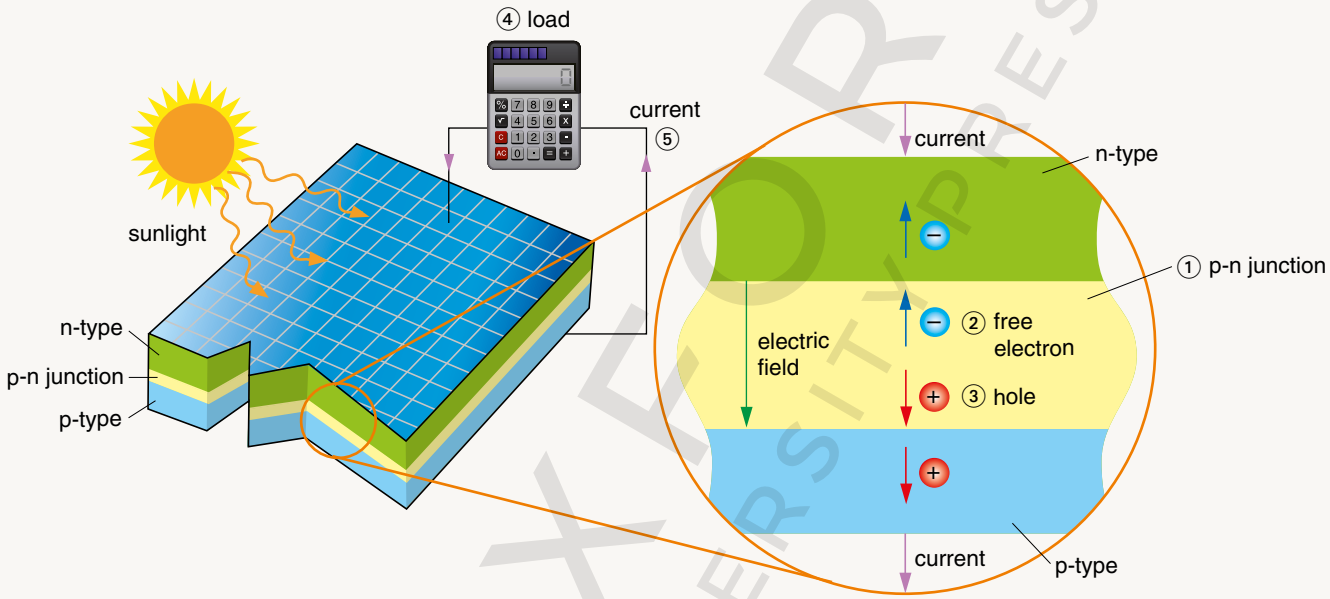
5 Solar power

- solar constant: 1366 W m^{-2} (just outside the Earth's atmosphere)
- when sunlight incident at an angle θ on a surface just outside the Earth's atmosphere,

  total power received per unit area = solar constant × cos θ



- working principle of solar cell:



① At the p-n junction, electrons in atoms absorb sunlight and become free electrons, leaving holes behind.

② The free electrons move against the direction of the electric field towards the n-side and migrate to the top surface of the cell.

③ The holes move along the direction of the electric field towards the p-side and migrate to the bottom surface of the cell.

④ If the cell is connected to a load, free electrons will flow from the n-side to the p-side via the load, and rejoin the holes at the p-side.

⑤ A d.c. current flows from the p-side to the n-side via the load.

Q28 Exam report: Part (a)(i) was in general well answered. Some candidates failed to realise that the power of the Sun distributes evenly on a spherical surface according to the inverse square law. In (a)(ii), quite a number of the candidates were unable to account for the energy loss of the solar radiation power through the atmosphere. Not many managed to describe the energy conversations in (b)(i) correctly. Some wrongly thought that it was electrical energy instead of chemical energy being stored in the battery while a few believed that heat or heat and light energy were being converted by the solar panel. Most managed to find the efficiency of the solar panel in (b)(ii). Candidates' performance in (b)(iii) was fair. Some candidates did not realise that the capacity '100 Ah 12 V' of the battery actually gives the maximum energy 1.2 kW h which can be stored. Many failed to get the correct answer as they wrongly multiplied this energy by the time of charging or made mistakes in the charging efficiency.

Key terms

1 binding energy 結合能 p.133
2 control rod 控制棒 p.141
3 coolant 冷卻劑 p.140
4 fossil fuel 化石燃料 p.132
5 global warming 全球暖化 p.167
6 greenhouse effect 温室效應 p.167
7 greenhouse gas 温室氣體 p.167
8 hydroelectric power 水力發電 p.144
9 moderator 減速劑 p.140
10 non-renewable energy source 不可再生能源 p.132
11 nuclear power 核能 p.132
12 photovoltaic cell 光伏電池 p.156
13 pressurized water reactor 壓水式反應堆 p.139
14 pumped hydroelectric storage system 抽水蓄能系統 p.150
15 renewable energy source 可再生能源 p.132
16 solar cell 太陽能電池 p.156
17 solar constant 太陽常數 p.152
18 solar panel 太陽能電池板 p.156
19 solar power 太陽能 p.144
20 wind power 風能 p.144
21 wind turbine 風力渦輪機 p.144

Revision 4

Take 1 u = 1.661×10^{-27} kg = 931 MeV, $e = 1.60 \times 10^{-19}$ C.

Multiple-choice questions

1 4.2 Which of the following is **not** a reason why the amount of greenhouse gases in the atmosphere is increasing fast?

A Growth in world's population
B Use of fossil fuels
C Use of nuclear power
D Deforestation

***2** 4.1 Which of the following statements about non-renewable energy sources are correct?

(1) Greenhouse gases are produced when using these energy sources to generate electricity.
(2) They will be depleted.
(3) They are the main energy sources used to generate electricity globally.

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

***3** 4.2 Which of the following statements about the greenhouse effect is/are correct?

(1) It is essential to human beings.
(2) It is produced entirely by human.
(3) It is affected by the amount of carbon dioxide in the atmosphere.

A (1) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

***4** 4.1 Mars is 2.28×10^{11} m away from the Sun. If the total power radiated by the Sun is 3.85×10^{26} W, what is the power of solar radiation per unit area reaching the atmosphere of Mars?

A 589 W m^{-2}
B 1770 W m^{-2}
C 2360 W m^{-2}
D 7410 W m^{-2}

***5** 4.1 Energy is released in a nuclear reaction. Which of the following statements about the changes in the reaction must be correct?

(1) The total mass of the particles increases.
(2) The number of particles increases.
(3) The total binding energy of the particles increases.

A (1) only
B (2) only
C (3) only
D (2) and (3) only

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★6 4.1 In a hydroelectric power station, 3.30×10^{10} kg of water flows through a vertical distance of 200 m in 2 hours. Estimate the efficiency of the station if its average output power is 7 GW.

A 17.8% B 56.1%

C 77.8% D 87.5%

★7 4.1 The binding energy of a $^{7}_{3}$Li nucleus is 39.244 037 MeV. What is the mass of this nucleus?

Given: mass of proton = 1.007 276 u
mass of neutron = 1.008 665 u

A 7.014 u

B 7.036 u

C 7.050 u

D 7.063 u

★8 4.1 Which of the following about hydroelectric power is correct?

A Its output is unstable.

B Its dam severely affects the natural habitat.

C It is suitable for any place.

D It has a low efficiency in generating electricity.

★★9 4.1 The blades of a wind turbine are of length 25 m. The figure below shows how the wind speed varies with time (Fig a).

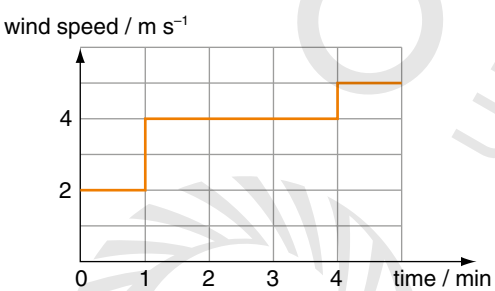


Fig a

Suppose the wind always blows normally onto the blades. What is the average power output of the turbine, in W, in the first 4 minutes? Take the efficiency of the turbine as 20%, and the air density in kg m^{-3} as ρ.

A $1070\pi\rho$

B $2250\pi\rho$

C $2680\pi\rho$

D $3125\pi\rho$

▶ Refer Eg4 (p.146)

★★10 4.1 A solar panel is illuminated by sunlight which is incident at an angle of 25° to the normal of the panel (Fig b) and the output power is 60 W.

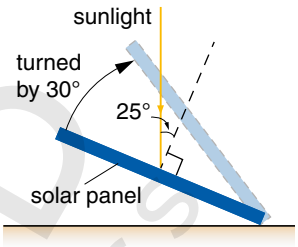


Fig b

What is the output power if the panel is turned by 30° clockwise as shown?

A 34 W

B 38 W

C 52 W

D 104 W

▶ Refer Eg7 (p.154)

11 HKDSE 2014 Paper 2 Q3.8 4.1

Under normal operation, which of the following statements about a pressurized water reactor (PWR) of a nuclear power plant is/are correct?

(1) The coolant which carries energy away from the reactor is radioactive.

(2) The steam that drives the turbine is radioactive.

(3) The cooling water discharged into the sea from the nuclear power plant contains some radioactive substances of the reactor.

A (1) only (39%)

B (3) only

C (1) and (2) only

D (2) and (3) only

12 HKDSE 2015 Paper 2 Q3.8 4.1

What is the function of the **moderator** in a nuclear fission reactor?

A It slows down neutrons and this helps increase the rate of nuclear fission. (32%)

B It slows down neutrons and this helps reduce the rate of nuclear fission.

C It absorbs neutrons and this helps reduce the rate of nuclear fission.

D It generates neutrons for nuclear fission.

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13 HKDSE 2016 Paper 2 Q3.4
4.1

Which statement about a solar cell is **incorrect**?

A When sunlight shines on a solar cell, some electrons in the semiconductor are excited to be free electrons.

B When a solar cell delivers power, current only flows in the interface at the p-n junction. (48%)

C The output voltage of a solar cell remains more or less unchanged when the incident light intensity increases.

D The efficiency of typical solar cells is about 10% to 20%.

14 HKDSE 2017 Paper 2 Q3.8
4.1

Energy is released in the following nuclear fission of uranium-235.

$$^{235}_{92}\text{U} + {}^{1}_{0}\text{n} \rightarrow {}^{94}_{40}\text{Zr} + {}^{139}_{52}\text{Te} + 3{}^{1}_{0}\text{n}$$

Which of the following statements concerning the reaction is/are correct?

(1) The rate of the reaction can be controlled by absorbing some of the neutrons produced.

(2) Mass is conserved in the reaction.

(3) The binding energy per nucleon of $^{235}_{92}\text{U}$ is higher than that of $^{94}_{40}\text{Zr}$ or $^{139}_{52}\text{Te}$.

A (1) only (52%) B (3) only

C (1) and (2) only D (2) and (3) only

15 HKDSE 2019 Paper 2 Q3.8
4.1

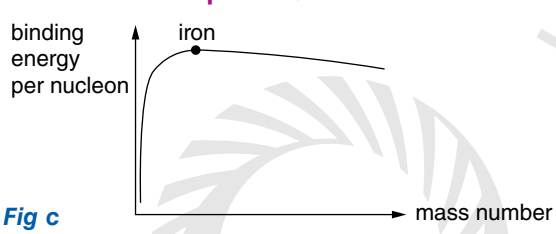


Fig c

Which of the following can be directly inferred from the above binding energy curve (Fig c)?

(1) The negative slope of the curve indicates that the total mass of the nuclides produced after fission is larger than the mass of the heavy nucleus before fission.

(2) The positive steep slope of the curve characterises that nuclear fusion in general produces much more energy per nucleon than nuclear fission.

(3) Iron being at the peak of the curve indicates that this element has the most stable atomic nucleus.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only (52%)

D (1), (2) and (3)

16 HKDSE 2021 Paper 2 Q3.7
4.1

Figure d shows a hydroelectric power plant.

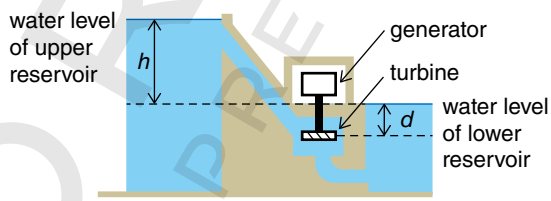


Fig d

Which factors below can affect the maximum power output of the plant?

(1) The height difference between the water levels in the upper and lower reservoirs, *h*.

(2) The distance between the turbine and the water level of the lower reservoir, *d*.

(3) The rate of water flowing through the turbine.

A (1) and (2) only B (1) and (3) only (50%)

C (2) and (3) only D (1), (2) and (3)

Conventional questions

*17 Misc The energy sources used in a country for generating electricity are shown in Table a.

	Coal	Gas	Nuclear fuel
Main products after use	carbon dioxide, water, sulphur dioxide	carbon dioxide, water	barium, krypton
Known world reserves	200 years	65 years	>500 years
Efficiency of electricity production	40%	50%	35%

Table a

(a) (i) Give two advantages of burning gas rather than coal in power stations. (2 marks)

(ii) Give one advantage of burning coal rather than gas in power stations. (1 mark)

(b) Both coal and gas produce carbon dioxide when they are used to generate electricity. Discuss the effect on the environment of the release of carbon dioxide. (2 marks)

(c) Suggest two reasons why the country is still using nuclear power even it has the lowest efficiency among the three methods of electricity production. (2 marks)

★**18** 4.1 Figure e shows schematically the structure of a nuclear power station employing a pressurized water reactor.

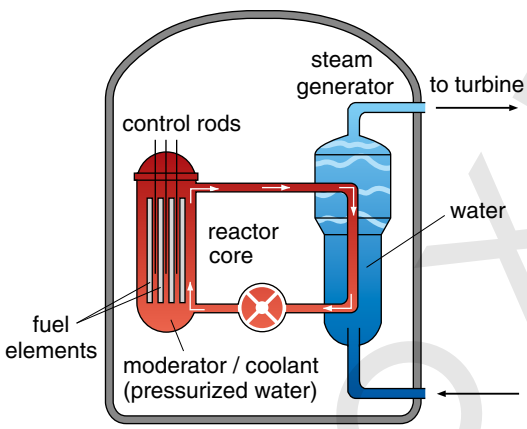


Fig e

(a) Explain the function of the moderator. (2 marks)

(b) Explain why pressurized water is pumped into the reactor core from the bottom and leaves it at the top. (2 marks)

(c) Why is the pressurized water prevented from mixing with the water in the steam generator? (1 mark)

★**19** 4.1 Solar panels are installed on the rooftop of a building (Fig f).



Fig f

Suppose the total area of the panels is 50 m^2. The efficiency of the panels is 19%. The solar constant is 1366 W m^{-2} and 30% of the solar power is absorbed by the atmosphere.

(a) Find the power output of the panels when they are illuminated normally by sunlight. (2 marks)

(b) Suppose that the panels are illuminated normally by sunlight for 9 hours each day. The cost of electricity is \$1.1 per kW h. How much can this system save every year? (2 marks)

★**20** 4.1 (a) What is the meaning of the binding energy of a nucleus? (1 mark)

(b) Sketch a graph to show how the binding energy per nucleon varies with the mass number. Indicate on the graph the positions of H-2, Fe-56 and U-238. (3 marks)

(c) Explain briefly why Fe-56 is unlikely to undergo fission or fusion. (1 mark)

★**21** 4.1 A small wind turbine is installed on the rooftop of a house to supplement the conventional power supply (Fig g). The blades of the turbine are 35 cm long. The density of air is 1.2 kg m^{-3}.

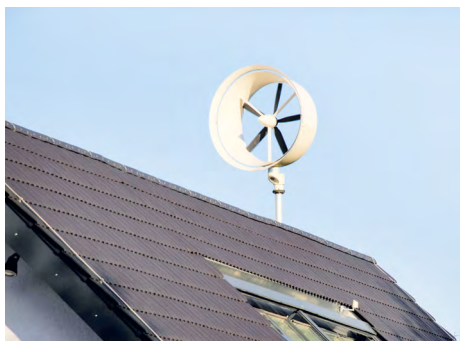


Fig g

(a) A wind of speed 8 m s^{-1} is blowing normal to the turbine and the overall efficiency of the turbine is 30%. Find the power generated by the turbine. (2 marks)

(b) Suppose that the cost of installing this system is \$5000. The cost of electricity is \$1.1 per kW h. Estimate the payback time of the system. (2 marks)

(c) Wind power is seldom used as the only power source. Give a reason for this. (1 mark)

★ **22** 4.1 A small island is situated far away from the mainland. The islanders require an electricity supply and suggest that wind turbines could be used together with natural gas powered generators.

(a) State the energy conversions in generating electricity using wind power. (1 mark)

(b) Suggest two advantages of using wind power on this island. (2 marks)

(c) Why is wind power used together with another energy source? (1 mark)

(d) There is no mountain and no large river on the island. Suggest another method to generate electricity on the island. (1 mark)

★ **23** 4.1 Albert's home has an average daily energy consumption of 30 kW h. Albert has installed some solar panels (Fig h) so that his home is entirely powered by solar power.



Fig h

(a) Describe the working principle of a solar cell. (4 marks)

(b) The efficiency of the solar panels is 15%. Suppose the Sun shines for 10 hours each day and the average power of the solar radiation per unit area received by the solar panels is 800 W m^{-2} during this time period.

(i) What is the minimum area of solar panels required to meet Albert's daily energy consumption? (2 marks)

(ii) In practice, an area larger than the result in (i) is needed. Give a reason for this. (1 mark)

(iii) What device, other than solar panels and cables, is needed to maintain the electricity supply at Albert's home? (1 mark)

★ **24** 4.1 A satellite at the outer surface of the Earth's atmosphere has solar panels of total area 1.2×10^{-4} km^2. The solar constant is 1366 W m^{-2}.

(a) Write down the energy conversion performed by a solar panel. (1 mark)

(b) Find the maximum solar power received by the solar panels. (2 marks)

(c) The efficiency of the solar panels is 20%. Find the maximum power output of the panels. (1 mark)

(d) The solar panels can be rotated. Write down an advantage of this design. (1 mark)

(e) The satellite is equipped with batteries which can be charged by the solar panels. Write down an advantage of this design. (1 mark)

★ **25** 4.2 The electrical energy used by a building is supplied by a coal-fired power station.

(a) State two effects to the environment caused by the extraction of coal. (2 marks)

(b) How does the conversion of the energy stored in coal to electrical energy contribute to global warming? (2 marks)

(c) State two major usages of electricity in Hong Kong. (2 marks)

★★ **26** 4.1 In a pumped hydroelectric storage system (Fig i), the vertical distance between the water levels of the upper and lower reservoirs is 120 m and the turbine is below the water level of the lower reservoir.

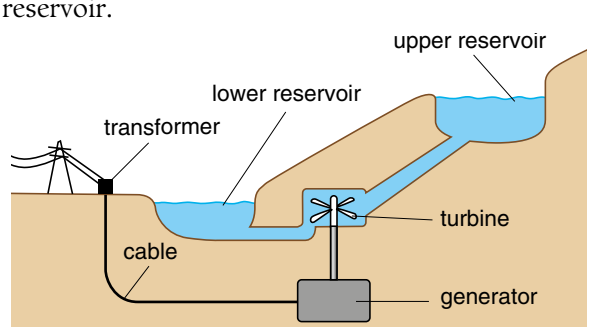


Fig i

(a) At night, electricity from a conventional power station is used to pump water from the lower reservoir to the upper reservoir. The efficiency in pumping water is 85%.

(i) State the energy conversion during this process. (1 mark)

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(ii) If 2.8×10^8 kg of water is pumped to the upper reservoir in 1 hour, find the average input power provided to the system. (3 marks)

(b) During daytime, water from the upper reservoir flows through the turbine to generate electricity. Suggest why this system stores energy at night and generates electricity during daytime. (2 marks)

(c) A solar power station can work with a pumped hydroelectric storage system. What is the advantage in doing this? (2 marks)

Refer p.150–151

See p.174 for the Exam Report.

27 HKDSE 2016 Paper 2 Q.3

4.1

Figure j shows a wind power station backed up by a pumped hydroelectric storage system via a control centre. Excess electrical power from the wind farm can be used to pump water from the lower reservoir to the upper one during off-peak hours. During peak hours, water runs down from the upper reservoir to drive the turbine and generator to produce electricity.

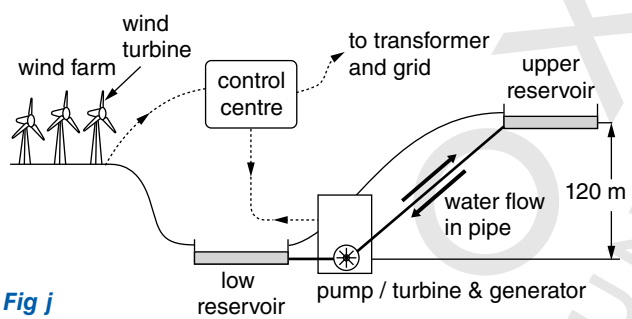


Fig j

A rotor blade of each wind turbine is 30 m long. Each turbine can be automatically controlled so that the blades' rotational plane is always normal to the wind direction. The graph in Figure k shows how the electrical power output from each turbine varies with wind speed.

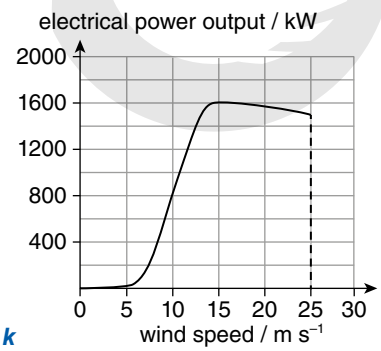


Fig k

(a) (i) State the reason why practically there is no power output from the turbine when the wind speed is

(1) lower than 5 m s^{-1}; and

(2) higher than 25 m s^{-1}.

(2 marks)

(ii) The turbine attains maximum power output when the wind speed is 15 m s^{-1}. Find the efficiency of the wind turbine in converting wind energy to electrical energy at such wind speed. Given: density of air = 1.23 kg m^{-3} (2 marks)

(b) There are 50 wind turbines in the wind farm for supplying electricity to a town. Figure l shows the variation of the town's electrical power demand with time of a day.

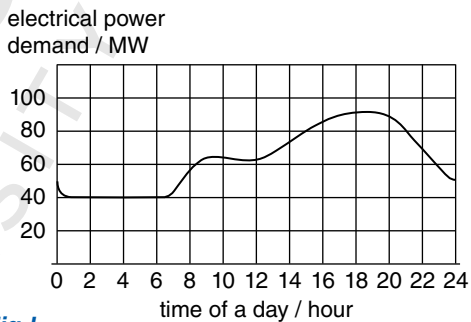


Fig l

(i) Using the two graphs given, find the lowest wind speed needed to meet the town's minimum demand of electrical power in a day without using the pumped hydroelectric storage system. (2 marks)

(ii) Suppose that on a certain day the wind speed is always 15 m s^{-1}.

(1) Estimate the total power output of the wind farm. Hence state the time period within which the pumped hydroelectric storage system needs to generate electricity for the town. (2 marks)

(2) During the period of minimum demand of electrical power by the town, at what flow rate, in kg s^{-1}, is water in the lower reservoir pumped back to the upper one at a vertical height of 120 m? The overall efficiency of the pump is 80%. (g = 9.81 m s^{-2}) (2 marks)

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See p.175 for the Exam Report.

28 4.1 **HKDSE 2019 Paper 2 Q.3**

(a) The total power of the Sun is about 3.86×10^{26} W, which radiates evenly in all direction. The Earth is at a mean distance of 1.50×10^{11} m from the Sun.

(i) Estimate the solar radiation power per unit area that can be obtained at the same distance of the Earth from the Sun. (2 marks)

(ii) State a reason why the maximum solar radiation power per unit area received on the Earth's surface normal to the Sun is only around 70% of that found in (a)(i). (1 mark)

(b) In the domestic energy storage system shown in the simplified schematic diagram (Fig m), energy from the Sun reaching a solar panel can be stored in a battery.

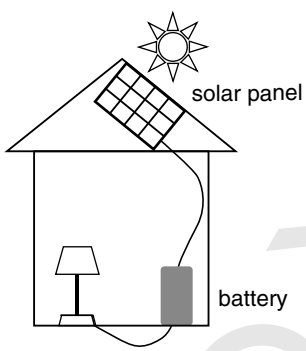


Fig m

The solar panel of area 1.65 m^2 is connected to the battery via a charger controller (not shown in Figure n). The solar panel delivers 300 W when it is normal to the Sun on a sunny day. Given: solar radiation power per unit area received on the Earth's surface = 1000 W m^{-2}

(i) Describe the energy conversions during charging in this domestic energy storage system. (2 marks)

(ii) Find the efficiency of the solar panel. (2 marks)

(iii) The capacity of the storage battery is '100 Ah 12 V'. How long would it take for the solar panel to fully charge the battery, which is completely discharged initially, if 20% energy loss occurs during charging? State one assumption in your calculation. (3 marks)

☆ Unit 1

29 Misc **CCEA GCSE Higher Tier 2019 Q2(a), (b)**

(a) Electrical energy can be obtained by burning fossil fuels.

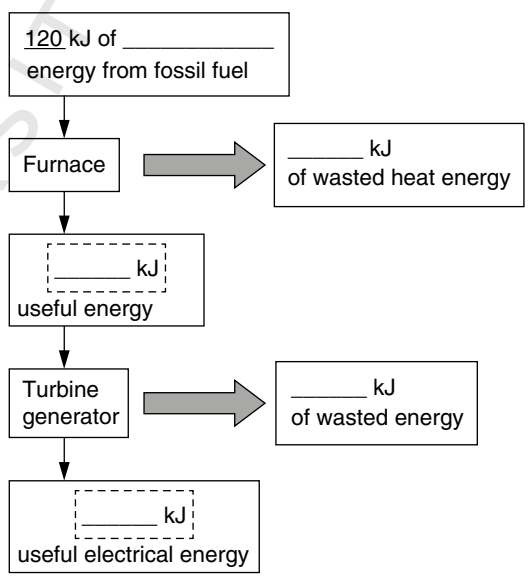
The simplified diagram below illustrates the energy changes that take place during this process.

For every 120 kJ of input energy, $\frac{2}{3}$ of it are wasted as heat from the furnace.

Of the remaining useful energy, 16 kJ are lost as heat and sound in the turbine. The remainder of the useful heat energy is converted to electrical energy.

(i) Complete the diagram below by adding the appropriate numbers to the boxes.

You should also name the type of input energy used in this process by adding the name to the first box.



(5 marks)

(ii) State, in full, the principle that allowed you to answer part (a)(i). (2 marks)

(b) Every second a different fossil fuel power station produces 840 MJ of electrical energy.

The efficiency of this power station is 0.35 (35%).

(i) Calculate the amount of fossil fuel energy this power station uses per second.

Show clearly how you get your answer, starting with the equation you plan to use. (4 marks)

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One of the major disadvantages of fossil fuel power stations is that they all produce gases which pollute the atmosphere.

(ii) State the common polluting gas produced by all fossil fuel power stations. (1 mark)

(iii) In what way is this gas particularly harmful to the global environment? (1 mark)

(iv) All fossil fuels are non-renewable. Explain what this means. (1 mark)

See p.184 for the Exam Report.

30 HKDSE 2020 Paper 2 Q.3

4.1

Pressurized water reactors constitute the majority of the world's nuclear power plants. In the reactor, energy is produced by the fission of uranium-235 nuclei ($^{235}_{92}U$). A typical fission reaction is as follows:

$$^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{144}_{56}Ba + {}^{90}_{36}Kr + 2{}^{1}_{0}n$$

(a) Referring to the binding energy curve in Figure n, explain why uranium-235 nuclei tend to undergo fission. (2 marks)

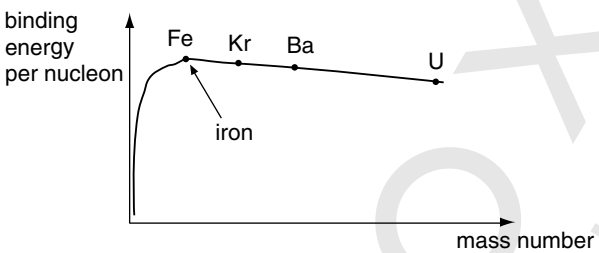


Fig n

(b) The binding energy of a uranium-235 nucleus is 1783 MeV.

(i) What does the above statement mean? (1 mark)

(ii) Find the energy released, in MeV, in the fission of a uranium-235 nucleus.

Given: Binding energy per nucleon of $^{144}_{56}Ba$ nucleus = 8.27 MeV per nucleon

Binding energy per nucleon of $^{90}_{36}Kr$ nucleus = 8.59 MeV per nucleon

(2 marks)

(c) (i) In the reactor of a nuclear power plant, a total energy of 1.30×10^{30} MeV would be released if all uranium-235 nuclei in the fuel rods have undergone fission. Given that the mean power output of the power plant is 500 MW and the efficiency in converting nuclear energy to electrical energy is 40%. Estimate the time, in years, for which the fuel rods can be used. (Take 1 year = 3.15×10^7 s) (2 marks)

(ii) State a reason why the fuel rods are usually replaced before the time estimated in (c)(i) has elapsed. (1 mark)

(d) Explain the role of the following in a fission reactor:

(i) moderator

(ii) control rods

(2 marks)

31 HKDSE 2022 Paper 2 Q.3

4.1

Mr. Chan plans to install a solar photovoltaic system on the roof of his house for generating electricity domestically.

(a) Each solar panel in the system has an area of 1.934 m^2. When it is normal to the Sun, the output voltage and current are 38 V and 10 A respectively on a clear day. Given: the maximum solar power per unit area reaching the Earth's surface is 1000 W m^{-2}

(i) Find the output power of the solar panel and estimate its efficiency. (2 marks)

(ii) In order to achieve a generation capacity of about 10 kW but not exceeding it, find the number of such solar panels to be installed and the corresponding minimum roof area required. (2 marks)

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(b) Mr. Chan can take part in the 'Feed-in Tariff (FiT) Scheme' by connecting his solar photovoltaic system to the power grid of the power company via an inverter as shown in Figure o. The renewable energy generated can then be sold to the power company.

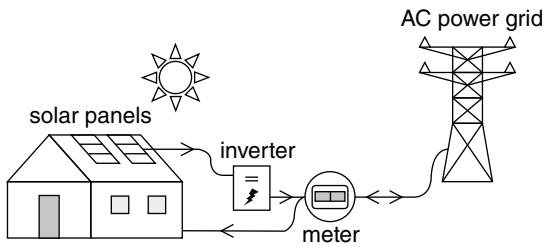


Fig o

(i) Why an inverter is needed to be installed between the solar panels and the AC power grid? (1 mark)

(ii) Throughout a year, Hong Kong has a daily average of 4.5 effective sunshine hours. Estimate the maximum energy, in kW h, that a solar photovoltaic system of 10 kW generation capacity can deliver in one year. (1 mark)

(iii) Suggest the main reason why the actual amount of electrical energy generated is far less than the value estimated in (b)(ii). (1 mark)

(iv) Given: the FiT Scheme offers a rate of $5 per kW h for a renewable energy system of capacity which does not exceed 10 kW and at most 10 000 kW h of the renewable energy generated can be sold to the power company per year. If the initial construction cost of such a system is $200 000 and the maintenance cost is $5000 per year, estimate the number of years it would take to recover the investment capital. (2 marks)

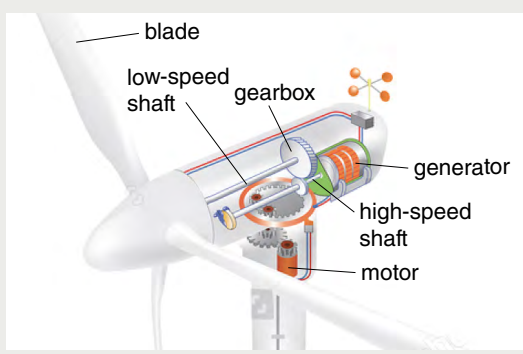
(c) State the advantage of solar photovoltaic systems over wind power systems for domestic generation. (1 mark)

Q31 Exam report: Part (a)(i) was well answered except for a few candidates who got input and output power mixed up. In (a)(ii), some candidates overlooked the stated generation capacity limit and casually rounded up the 26.3 solar panels found to 27. One-third of them knew the function of the inverter in (b)(i). Candidates performed well in (b)(ii). For the explanation in (b)(iii), many stated factors which had already been given indirectly in the question e.g. weather conditions or efficiency of solar panels. Candidates did well in the estimation in (b)(iv). Many answers regarding the comparison between solar and wind power systems were not based on domestic generation as required in (c).

Know more

Structure of a wind turbine (p.145)

When the wind blows and makes the blades of the turbine spin, the shaft connected to the generator rotates and electricity is generated by the process of electromagnetic induction. Usually, the turbine has gears which translate the slow rotation of the blade into a quicker rotation for generating electricity. Furthermore, the turbine can orient itself in the direction of the wind to maximize power generation.

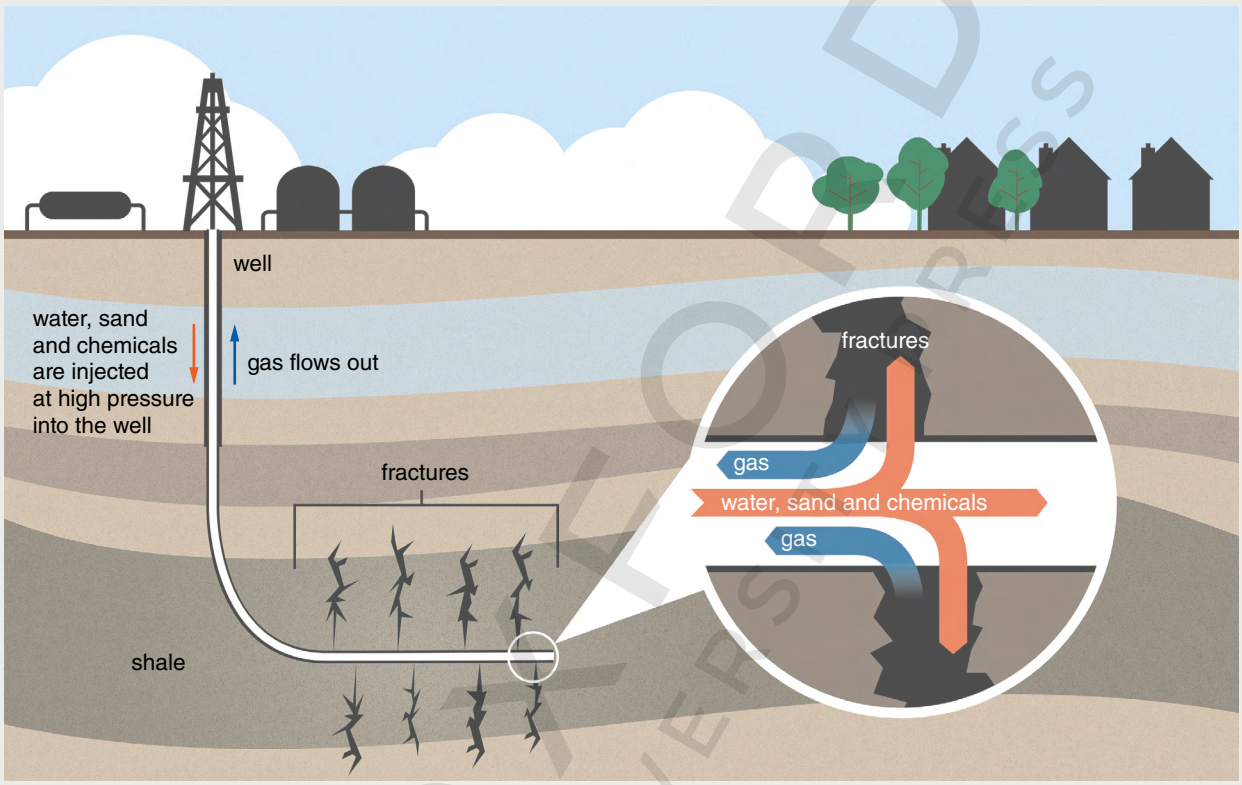


Q30 Exam report: Candidates' performance was only fair. In (a), not many were able to mention that an atom or a nucleus is more stable if it possesses a higher binding energy per nucleon. In (b), many candidates did not understand that 'binding energy' represents the 'energy required to separate the nucleons completely'. Weaker candidates held that it was the amount of energy needed to 'bind' the nucleus from its constituent nucleons. For the calculation in (b)(ii), some incorrectly multiplied the binding energy per nucleon by the difference of mass number and atomic number. Less than half of the candidates obtained the correct answer in (c)(i). Many made mistakes in units conversion or in manipulating the efficiency of energy conversion. In (c)(ii), few were able to mention that a chain reaction could not be sustained when the concentration of U-235 in the fuel rods was too low. In (d), most candidates knew the function of moderator in a fission reactor though a few wrongly stated that 'electrons' or 'atoms' were to be slowed down. For those who pointed out that control rods were to absorb neutrons, some did not further explain that they were for controlling the rate of reaction.

Know more

Shale gas (p.132)

Shale gas is a natural gas trapped within *shale* formations. This gas is much harder to extract from the ground than conventional natural gas. Because of advances in technology, the extraction of shale gas has increased rapidly in recent years. There are many shale gas reserves all over the world, with the reserves in our country being the largest. Experts expect that shale gas will greatly increase the world's energy supply.



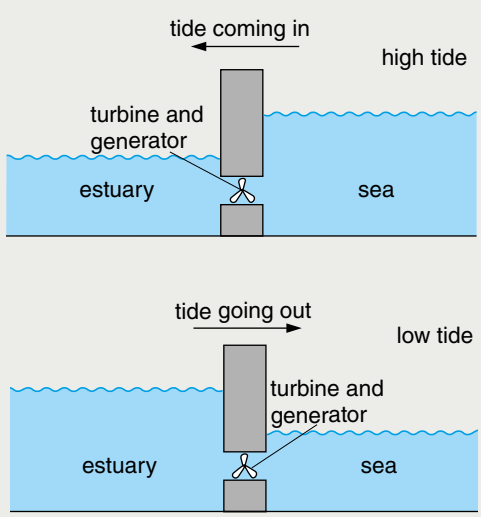
A method used to extract shale gas.

Tidal power (p.149)

Tidal power also makes use of water to generate electricity. The generators work like wind turbines built underwater to convert the kinetic energy of water from tidal currents to electrical energy.



A dam is built across a river estuary.



When the tide goes in and out, the water flows through tunnels in the dam and drives the generators.

shale 頁岩　　river estuary 河口

STSE

Renewable energy sources in Hong Kong (p.146)

Individual projects on renewable energy sources have been carried out since the 1980s.

Below are some of the largest projects launched in recent years. Can renewable energy sources be developed on a larger scale in Hong Kong? What are the constraints? Which kinds of renewable sources are most suitable for Hong Kong?

Lamma Winds

Lamma Winds, located at Lamma Island, is the only large-scale wind power station in Hong Kong. The wind turbine is 71 m high and the blades are 25 m long. It can generate 10^6 kW h of electricity each year.



Floating Solar Power System at Plover Cove Reservoir

352 solar panels have been installed as a floating platform on the surface of Plover Cove Reservoir, covering an area of about 1100 m^2. It can generate 120 000 kW h of electricity each year.



Hydropower system in treatment works

This hydropower system has been installed at Tuen Mun Water Treatment Works and Sha Tin Treatment Works. When water in the reservoir flows to the water tank at a lower level in the treatment works, it turns the turbine and generates electricity.

